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Multi-objective optimization of hybrid renewable energy systems with green hydrogen integration and hybrid storage strategies

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ABSTRACT

This study proposes a hybrid renewable energy system (HRES) that integrates photovoltaic panels (PVs), wind turbines (WTs), and continuous green hydrogen production via reformers. To enhance system reliability and efficiency, hybrid storage solutions, including hydrogen tanks and batteries, are incorporated. An integrated size and design optimization algorithm (i-NSGA-II), based on the Non-dominated Sorting Genetic Algorithm-II (NSGA-II), is proposed, combining the system design optimization with an advanced power management strategy (PMS). The proposed approach is validated through comparison with multi-objective particle swarm optimization (MOPSO). In the hydrogen storage scenario, the minimum annualized cost of system (ACS) solutions achieved a 10% reduction in ACS, the minimum loss of power supply probability (LPSP) solutions showed a 78% decrease in LPSP, and the minimum percentage of energy waste (PEW) solutions resulted in an 83% reduction in PEW compared to MOPSO. In the hybrid storage scenario, the results showed a 9% reduction in ACS and an 85% decrease in LPSP for the corresponding objective functions compared to MOPSO. For both minimum PEW solutions in the hybrid storage scenario, the PEW value was 0%, achieving 100% system efficiency. Simulations conducted in three regions demonstrated the system's adaptability to varying climatic and regional conditions. A comprehensive sensitivity analysis reveals the impacts of cost variations, storage types, and capacities on system performance, providing insights into the trade-offs involved in optimizing HRES.

1. Introduction

The depletion of fossil fuel reserves and their significant environmental consequences have heightened the demand for sustainable energy alternatives. Solar and wind power have emerged as leading candidates due to their availability and environmental benefits. [1]. However, it is well-known that when used alone, the unpredictable and intermittent nature of these resources often fails to sufficiently meet the cost and reliability expectations for providing energy to either a single residence or larger-scale systems. To address this challenge and better harness these renewable resources, hybrid energy systems have been developed, incorporating them together along with power backup units [2–4].

Building on the development of these hybrid energy systems, a critical next step is to ensure their long-term viability through a balanced optimization of various performance indicators. Studies have shown that for these systems to be economically feasible, factors such as system reliability, CO_2 emissions, and energy usage efficiency need to be optimized, not just the system size. These findings underscore the importance of employing advanced multi-objective optimization strategies like the Pareto optimization. The Pareto optimal solution approach allows decision-makers to prioritize solution areas according to their preferences [5,6]. The solution set obtained includes all optimal solutions that cannot dominate each other, known as the Pareto Front, a concept introduced by Vilfredo Pareto [7,8]. These solutions cannot be improved in one objective without simultaneously worsening another, thus they are not dominated by any other member within the population [9].

Given its advantages, Pareto-based optimization has been widely applied in HRES design. In this context, algorithms such as NSGA-II and MOPSO are frequently used to balance multiple objectives. For instance, Baghaee et al. proposed a novel method for minimizing the annual cost, expected energy loss, and expected load loss, of a microgrid system consisting of hydrogen storage, PV, wind, and fuel cells. This method is

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composed of an approximate method to reduce computation time and MOPSO algorithm. Simulation results demonstrate the impact of component failures on the cost and reliability of the system [10]. In another study on Pareto optimization, Ansari [11] focused on minimizing TNPC and LPSP using the NSGA-II to provide electricity to a healthcare center with a HES consisting of hydrogen storage, PVs, and a diesel generator (DG). To select the best solution from the Pareto-optimal set, Ansari employed a multi-criteria decision-making approach based on the Analytic Hierarchy Process (AHP) and the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS). Additionally, a sensitivity analysis was performed, considering the impacts of fuel and storage costs. Ma et al. [12] introduced a Hypervolume-enhanced NSGA-III algorithm for the configuration of HRES and evaluated its performance against other Pareto optimization techniques like MOPSO. Their findings highlighted the hydrogen subsystem's role in improving system reliability and sustainability, alongside the algorithm's exceptional Pareto performance. Studies that propose a Pareto-based multi-objective optimization method for HRESs can also be examined in Refs. [13–17].

HRES composed entirely of renewable sources, along with diverse storage technologies, are increasingly being adopted for zero-emission and sustainable power solutions. In this context, the usage of hydrogen in HRES, as both a renewable energy source and a storage tool, is gradually increasing due to its clean, sustainable, and environmentally friendly attributes [18].

In recent years, the production of hydrogen in HRES by splitting water into hydrogen and oxygen atoms in electrolyzers using the system's excess energy has become widespread. A review of the literature reveals that optimization studies on HRES using hydrogen obtained through electrolysis as a storage medium are divided into two categories: systems entirely based on hydrogen storage and systems that also utilize battery storage and/or other storage techniques. Focusing specifically on studies that utilize hydrogen storage alone, several optimization approaches have been proposed. Kalinci et al. [19] presented an HRES model using hydrogen storage for Bozcaada Island in Türkiye. Using the Hybrid Optimization Model for Electric Renewable (HOMER), they determined the optimal system size providing minimum NPC and cost of energy (COE) through scenarios comparing only wind turbines and wind turbine/PV hybrid systems. They found that integrating PV arrays with wind turbines significantly reduced the NPC, thereby affecting the sizes of the hydrogen tank (HT) and other equipment, as well as the COE. Samy et al. [20] developed the Flower Pollination Algorithm (FPA) for size optimization of PV and fuel cell integrated energy systems for remote and isolated regions in Egypt based on total annual cost. MATLAB™ simulations demonstrated that FPA is faster and more efficient compared to other optimization techniques. The optimal hybrid system configuration resulted in a total net present value of $3,244,897 and an LCOE of $0.334/kWh, with LPSP constrained to 1.52%. In their subsequent study, Samy et al. [21] present combinations of hybrid systems: PV/FC, PV/WT/FC, and WT/FC. They compared the results obtained from the FPA with those from the Shuffled Frog Leaping Algorithm (SFLA) and the Particle Swarm Optimization (PSO) techniques. They demonstrated that the PV/wind/FC combination outperformed others with an LCOE of $0.47/kWh. Ceylan and Devrim [22] designed an HRES to provide electricity and heat to a greenhouse in Şanlıurfa, Türkiye, and optimized it using MATLAB/Simulink. In this system which consists of PV, electrolyzers, and Proton Exchange Membrane Fuel Cells (PEMFC), the heat generated by the PEMFCs was utilized to heat the greenhouse through micro-cogeneration application. They calculated the Levelized Cost of Energy (LCE) as $0.117/kWh for this system.

The integration of hydrogen storage with alternative energy storage technologies, such as batteries, into HRES has garnered increasing attention in recent years [23]. Numerous studies have explored the single or multi-objective optimization of these systems. For instance, Sharafi and ElMekkawy [24] developed a novel approach that combines the ε-constraint method and PSO for the optimal design of a HRES consisting of WTs, PV, DG, batteries, and hydrogen storage. Their model simultaneously considers various renewable sources and storage devices, aiming to minimize the total cost, unmet load, and fuel emissions of the system. Altun and Kılıç [25] used a multi-objective optimization method based on the Taguchi approach to minimize the levelized energy cost and CO_2 emissions of grid-independent HRESs. They used battery banks or hydrogen systems as energy backup units. Medghalchi and Taylan [26] aimed to minimize energy costs across different scenarios by integrating PV, WT with hydrogen or battery storage options. For this purpose, they proposed a hybrid optimization system using PSO and Genetic Algorithm. Their findings indicate that the PV-wind hybrid system is the most effective with the lowest cost. Al-Quraan and Athamnah [27] conducted size and energy management optimization for a grid-independent HRES comprising WT and PV with battery and hydrogen backup systems. They proposed a three-stage optimization model consisting of nested loops to maximize system efficiency. They achieved an average efficiency of 83.53% while minimizing the total capital cost to $127,400 and the operating cost to $67. Another study that combines battery and hydrogen backup systems was conducted by Trapani et al. [28]. They utilized wind and solar energy as primary sources in a fully reliable uninterrupted HRES and minimized the LCOE using PSO. Song et al. [29] designed an off-grid HRES that utilizes solar energy, hydrogen, and retired batteries to provide power and heat. To optimize cost, improve reliability and minimize waste, they developed an NSGA-II algorithm that integrates the Transient System Simulation Program (TRNSYS) and MATLAB. Bade et al. [30] conducted the optimal design of a HRES integrating WTs, PV systems, biomass, battery, and hydrogen storage for a residential load, while considering technical, economic, environmental, and social performance indicators. They employed the hybrid genetic algorithm-particle swarm optimization (GAPSO) algorithm as a multi-objective optimization approach. The results of their study demonstrated that GAPSO achieved a cost-effectiveness improvement of 2.17% compared to GA and 0.01% compared to PSO.

In addition to integrating hydrogen and battery storage, studies incorporating various alternative storage techniques into HRES have also been conducted. In their research, Bryan et al. [31] devised an HRES incorporating thermal and hydrogen storage alongside solar, wind, and nuclear energy to fulfill all of Texas's electricity requirements. Even though the Levelized Cost of Energy (LCOE) obtained through their proposed optimization algorithm surpasses that of the existing system, they envisaged a system capable of generating carbon-free and green hydrogen. Guo and Niu [32] developed an optimization approach that integrates both single and multi-objective optimization for standalone HRESs integrated with solar and wind energy, batteries, a hydrogen energy storage system, and a thermal energy storage system. In the first stage, LCOE and LPSP were evaluated as objective functions within the multi-objective optimization framework. In the second stage, the optimal LPSP values obtained were used as constraints, and the LCOE was further optimized. The results demonstrated that lower LCOE values were achieved compared to those obtained through multi-objective optimization alone, while maintaining the same LPSP values. Jafarian et al. [33] conducted a study using TRNSYS simulation, response surface methodology, and life cycle assessment to highlight the trade-offs in multi-objective optimization for energy storage capacity, renewable fraction, and storage types. They assessed the impact of different energy storage types, such as latent energy storage, hydrogen storage, and battery storage, on optimal design. In another study, Li et al. [34] proposed a collaborative framework between integrated energy systems and hydrogen-natural gas energy storage system to enhance sustainability and cost optimization. By employing a MOPSO method, the study introduces advanced strategies for improving energy cooperation and reducing costs. The results demonstrate significant profitability gains and environmental advantages through better resource management.

Another method of obtaining hydrogen is the reforming of biogas derived from biomass. Biogas produced in anaerobic digesters designed

for biological waste treatment and in landfill gas recovery facilities has competitive advantages over natural gas, such as being a renewable energy source and reducing emissions by preventing methane release into the atmosphere [35]. Additionally, while CO_2 emissions from natural gas reforming contribute to global warming, CO_2 emissions from biogas reforming do not have a negative impact [35–37]. Therefore, biogas could serve as an alternative feedstock to natural gas in traditional steam methane reforming technologies. Studies utilizing hydrogen obtained by this method within an optimization framework in HRESs are rare in literature. Hakimi and Tafreshi [38] used a fuel cell powered by hydrogen obtained from a reformer, along with WTs and a HT for storing excess energy, in an off-grid HRES. They utilized the PSO algorithm for the system's size optimization. Tezer and Yaman [9] developed an NSGA–II–based algorithm for off-grid HRESs that includes ACS and LPSP minimization. Their algorithm, which considers the LPSP value and waste energy ratio as constraints, presents a PMS that maximizes hydrogen utilization by balancing these values. Additionally, they introduced an innovative design that incorporates hydrogen as an uninterruptible power supply into the hybrid system optimization. HassanzadehFard et al. [39] utilized two sources, methane produced from wastes and purchased natural gas, for hydrogen production via a reformer in a WT-PV-BESS-FC (reformer-fed) hybrid system. The minimum NPC value of the system was calculated under a specific LPSP constraint, where the reformer was not included in the size optimization.

In the reviewed and accessible literature, it is observed that hydrogen production in HRESs is often carried out using an electrolyzer to store excess energy. System elements that produce hydrogen from biomass, such as reformers, appear to be rarely used and are not included in size optimization. Although many systems integrate hydrogen and battery storage, such an integration has not been found in the absence of an electrolyzer.

To further emphasize the contributions and differences of the proposed system, a comparison is made with the systems in the literature. Table 1 compares the optimized system configurations and associated objective functions in various studies with the proposed systems for Scenario 1 and Scenario 2, highlighting the differences with the developed system.

The main objective of this study is to address the limitations identified in the literature regarding hydrogen production and storage by integrating uninterrupted hydrogen production from biomass sources via a reformer and its storage into HRES, and optimizing system cost, reliability, and efficiency with an innovative PMS. To achieve this, the i-NSGA-II algorithm was developed for the sizing and optimization of system components, and system performance was evaluated under different climatic conditions.

Key differences and contributions of the proposed approach compared to previous research.

- Unlike many studies that focus solely on hydrogen storage using electrolyzers, this research incorporates continuous hydrogen production through reformers, ensuring an uninterrupted hydrogen supply and enhancing system reliability and efficiency.
- Two distinct energy storage strategies are proposed: Scenario 1 utilizes two-source hydrogen storage, while Scenario 2 integrates hybrid battery and hydrogen storage, providing alternative solutions to optimize energy storage and reliability under varying conditions.
- The PMS optimizes energy flows by efficiently utilizing hydrogen and battery storage across different climatic conditions in Çeşme, Çanakkale, and Kahramanmaraş, showcasing trade-offs between cost, reliability, and efficiency.
- A comprehensive system design optimization framework is introduced, which not only integrates hydrogen production and hybrid storage systems but also optimizes their sizes and configurations based on specific conditions and interactions. This flexible approach ensures a more tailored and efficient system design, enhancing the practical applicability and flexibility of HRES configurations.
- The study employs i-NSGA-II and validates its performance through a comparison with the MOPSO, simultaneously optimizing cost, reliability, and efficiency.
- A comprehensive sensitivity analysis evaluates the effects of cost variations, storage types, and capacities on system performance, highlighting key factors that influence feasibility and providing insights into the trade-offs in optimizing HRES.

2. System description and methodology

2.1. Layout of the proposed hybrid system

The developed multi-objective algorithm can generate alternative solutions for any HRES composed entirely of renewable sources. However, as shown in Fig. 1, this study evaluates two different HRES configurations through two scenarios, highlighting and assessing various hydrogen production alternatives and storage options:

Fig. 1(a) illustrates the system configuration for Scenario 1. This system consists of WTs, PVs, electrolyzers that utilizes surplus energy in the system to produce hydrogen, an anaerobic reactor for biogas production from biomass, a reformer that ensures continuous hydrogen production by reforming the produced biogas, a fuel cell that converts the hydrogen produced by the electrolyzer and the reformer into electricity, and a HT for storing surplus hydrogen.

Fig. 1(b) represents the system configuration for Scenario 2. This configuration includes WTs, PVs, batteries for storing surplus energy in the system, an anaerobic reactor-reformer system that ensures

Table 1
Comparison of optimized system Configurations and objective functions in relevant literature.

	Optimized System Configuration									Objective Functions	
Ref.	PV	WT	Fuel Cell	Electrolyzer	Reformer	Hydrogen from waste	Hydrogen Storage	Battery Storage	System Cost	System Reliability	System Efficiency
[9]	✓	✓	✓	✓	✓	✓	✓		✓	✓	
[12]	✓	✓	✓	✓			✓	✓	✓	✓	✓
[20]	✓		✓	✓			✓		✓		
[26]	✓	✓	✓	✓			✓	✓	✓		
[30]	✓	✓	✓	✓		✓	✓	✓	✓		
[28]	✓	✓	✓	✓			✓		✓		
[29]	✓		✓	✓			✓	✓	✓	✓	✓
[38]		✓	✓	✓	Const.[a]	✓	✓		✓		
[39]	✓	✓	✓		Const.[a]	✓		✓	✓		
Proposed system (Scenario 1)	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓
Proposed system (Scenario 2)	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓

[a] Const.: 'Constant' refers to a component that has a fixed value and is excluded from the optimization process.

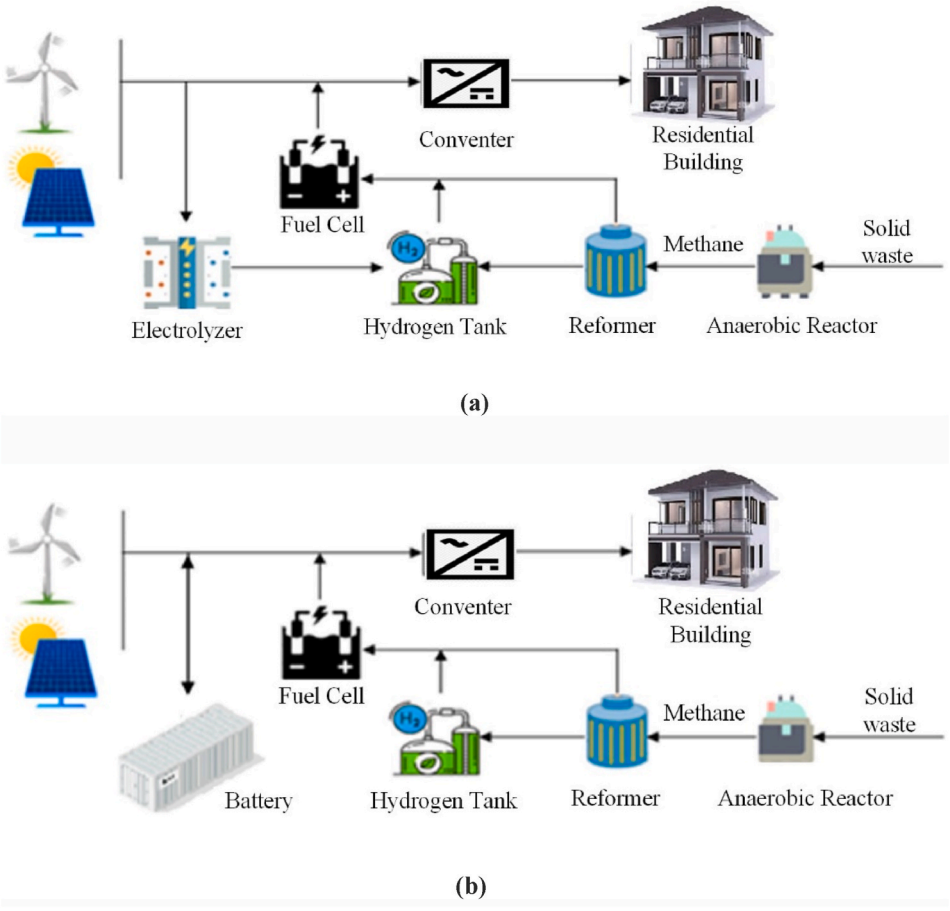


Fig. 1. Layout of the designed off-grid HRES (a) Scenario 1: HRES with hydrogen storage only (b) Scenario 2: HRES with both battery and hydrogen storage.

Table 2
Wind, solar and temperature data of Çanakkale, Çeşme and Kahramanmaraş.

	Total	Mean	Standard Deviation	Maximum	Minimum
Çanakkale wind speed data (m/s)	–	4.41	2.45	19.48	0.0456
Çeşme wind speed data (m/s)	–	4.39	2.91	10.00	0.0045
Kahramanmaraş wind speed data (m/s)	–	3.76	1.96	9.96	0.0000
Çanakkale solar radiation data (kWh/m^2)	1531.17	0.17	0.26	–	–
Çeşme solar radiation data (kWh/m^2)	1853.90	0.21	0.30	–	–
Kahramanmaraş solar radiation data (kWh/m^2)	1697.01	0.19	0.28	–	–
Çanakkale temperature data (°C)	–	15.25	7.82	26.16	4.79
Çeşme temperature data (°C)	–	17.78	5.19	25.34	11.09
Kahramanmaraş temperature data (°C)	–	11.82	8.86	24.48	−0.47

continuous hydrogen production, a fuel cell that generates electricity from hydrogen, and a HT for storing surplus hydrogen.

The two proposed scenarios differ in terms of hydrogen production and energy storage techniques. Scenario 1 stores two-source hydrogen, produced through the electrolyzer and reformer, in HT, while Scenario 2 uses HT to store hydrogen produced only by the reformer and employs a battery to store the surplus energy generated by the PV and WTs.

2.2. Renewable potential of the HRES

Solar radiation, temperature, and wind data were utilized for Çanakkale, located in the Marmara region, Çeşme district of İzmir in the Aegean region, and Kahramanmaraş province in the eastern Mediterranean region of Türkiye. Among them, Çeşme, although having less wind potential compared to Çanakkale, receives more solar radiation throughout the year. The annual total radiation value is 1854 kWh/m^2 for Çeşme, 1697 kWh/m^2 for Kahramanmaraş, and 1531 kWh/m^2 for Çanakkale. The daily average solar irradiance values, in kWh/m^2/day, calculated for Çeşme, Kahramanmaraş, and Çanakkale are 5.08, 4.65, and 4.19 respectively.

The values considered correspond to the coordinates 38 ° 18′ N and 26 ° 19′ E for Çeşme, 40 ° 9′ N, 26 ° 24′ E for Çanakkale and 38 ° 33′ N and 38 ° 24′ E for Kahramanmaraş. The data presented in Table 2 are statistical values calculated from the obtained HOMERPro® data for wind

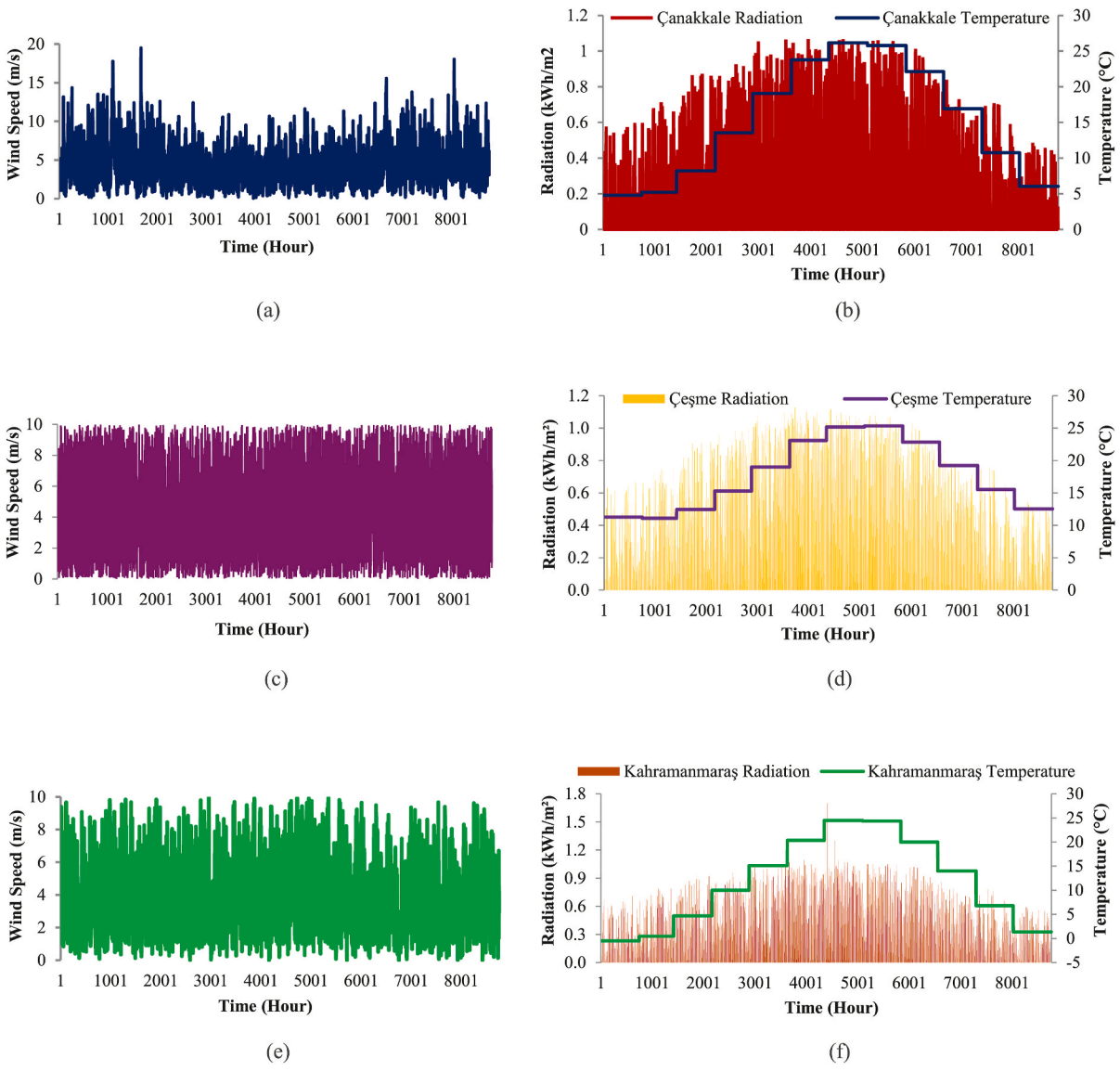


Fig. 2. Annual hourly data for three locations Wind speed data (a) Çanakkale (c) Çeşme (e) Kahramanmaraş Solar radiation and temperature data (b) Çanakkale (d) Çeşme (f) Kahramanmaraş.

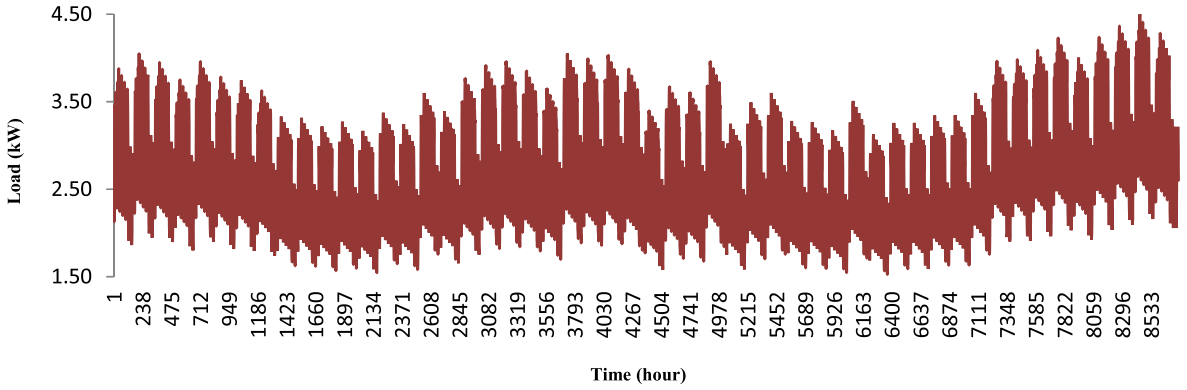


Fig. 3. Annual load data on hourly basis.

speed, solar radiation, and ambient temperature for each location, based on the coordinates mentioned. Fig. 2 depicts the variation of wind speed, solar radiation, and ambient temperature throughout the year for Çanakkale, Çeşme and Kahramanmaraş.

2.3. Load demand

To model typical household electricity consumption, the IEEE (Institute of Electrical and Electronics Engineers) RTS (Reliability Test System) annual hourly load data were utilized. The average and peak load values were set at 2.76 kW and 4.5 kW, respectively. Fig. 3 shows the hourly load data.

3. Mathematical estimation of HRES

3.1. PVs

The equation concerning the power output of a PV (P_{PV}) is presented in Eq. (1) [40,41]:

$$P_{PV} = P_{pv_rated} \times \frac{R}{R_{stc}} \times \left(1 + T_{co} \times \left(T - T_{ref}\right)\right) \tag{1}$$

In Eq. (1) above, P_{pv_rated} represents the nominal power under standard test conditions, R denotes the solar radiation (W/m^2), and R_{stc} signifies the solar radiation under standard test conditions (1000 W/m^2). Additionally, T_{ref} refers to the cell temperature under reference conditions, and T_{co} is the temperature coefficient. The cell temperature T is determined using the ambient temperature (T_{ort}) and R, as calculated with Eq. (2) [40]:

$$T = T_{ort} + R \times 0,0256 \tag{2}$$

3.2. Wind turbines

By utilizing the WT nominal power ($P_{wt_{rated}}$), WT hub height speed ($V_{hub}(t)$), cut-in wind speed (V_{cin}), rated wind speed (V_{rated}), and cut-out wind speed (V_{cout}), along with a fixed parameter k, the hourly output power of the WT ($P_{wt}(t)$) is calculated as follows [42,43]:

$$P_{wt}(t) = \begin{cases} 0 & 0 \le V_{hub} \le V_{cin} \quad and \quad V_{hub} > V_{cout} \\ \dfrac{V_{hub}(t)^k - V_{cin}{}^k}{V_{rated}{}^k - V_{cin}{}^k} \times P_{wt_{rated}} & V_{cin} \le V_{hub} < V_{rated} \\ P_{wt_{rated}} & V_{rated} \le V_{hub} < V_{cout} \end{cases} \tag{3}$$

To calculate $V_{hub}(t)$ at the hub height (h_{hub}), annual wind speed data measured hourly ($V_{ref}(t)$) at a reference height (h_{ref}) of 10 m is used:

$$V_{hub}(t) = V_{ref}(t) \times \frac{ln(h_{hub}/l_{sr})}{ln(h_{ref}/l_{sr})} \tag{4}$$

where l_{sr} is the surface roughness length (m), and its value for each WT type is retrieved from the corresponding table in the PMS algorithm.

3.3. Battery

In this study, one of the systems used to store excess energy from wind and solar sources and to provide energy in case of a deficit is batteries. The efficient use and longevity of batteries are ensured by the PMS, which prevents the charge and discharge states from exceeding the specified maximum and minimum values. In the designed HRES, if the excess energy exceeds the battery's maximum capacity, the PMS will ensure the excess energy is discharged as waste energy. In the case of an energy deficit, the batteries will first supply energy without dropping below its minimum capacity. If there is still insufficient energy, it will be provided by the fuel cell using hydrogen stored in HT, which is generated by the reformer.

$P_{exc_{pvwt}}$, at any time t, means the difference between the energy produced by PV (P_{pv}) and WTs (P_{wt}) and the energy demanded by the load (P_{load}). It is calculated using Eq. (5). P_{exc} is the difference between the load demand and the energy produced by the system and is calculated by Eq. (6). The battery charge and discharge status (P_{bat_ex}) at time t is calculated as shown in the following Eq. (7) and Eq. (8), respectively [16]:

$$P_{exc_{pvwt}} = P_{pv} + P_{wt} - P_{load}/e_{conv} \tag{5}$$

$$P_{exc} = P_{load} / e_{conv} - \left(P_{pv} + P_{wt} + P_{rfr}\right) \tag{6}$$

$$P_{bat_ex} = P_{bat(t-1)} + P_{exc_{pvwt}} \times e_{bat_ch} \tag{7}$$

$$P_{bat_ex} = P_{bat}(t-1) - P_{exc}/e_{bat_dch} \tag{8}$$

e_{bat_ch} and e_{bat_dch} represent the charging and discharging efficiencies of the battery, respectively. These values vary due to changes in the battery's charging and discharging currents. However, in this study, both values are assumed to be equal and set at 90%, as referenced in Ref. [44]. In addition, the self-discharge rate is not considered in the calculations of the battery's charge and discharge capacities.

DOD (depth of discharge) indicates the discharge limit of the battery. In this study, the DOD value is set at 90%. In this study, since the types as well as the numbers of system components are optimized, the maximum useable battery capacity and consequently the minimum capacity of the batteries are calculated using Eq. (9) and Eq. (10):

$$P_{bat_max} = P_{bat_r} \times n_{bat} \tag{9}$$

$$P_{bat_min} = P_{bat_max} \times (1 - dod) \tag{10}$$

The battery's capacity at time t should not exceed its minimum and maximum values. This constraint is ensured by the following Eq. (11) [6].

$$P_{bat_min} \le P_{bat}(t) \le P_{bat_max} \tag{11}$$

3.4. Electrolyzer

When the power produced by the WT and PV generators exceeds the demand, the excess energy is transferred to the electrolyzer and used to produce hydrogen for electricity generation in the fuel cell. By utilizing the excess power from the WTs and PVs (P_{exc_pvwt}), the electrolyzer efficiency (e_{ELC}), and the higher heating value of hydrogen (HHV_{H_2} of 39.4 kWh/kg), the amount of hydrogen produced (in kg) by the electrolyzer during the time interval t ($P_{ELC}(t)$) can be calculated as given in Eq. (12):

$$P_{ELC}(t) = \begin{cases} \dfrac{P_{exc_{pvwt}}(t) \times e_{ELC}}{HHV_{H_2}} & \text{if } P_{exc_{pvwt}}(t) \le P_{ELC_{rated}} \times n_{ELC} \\ \dfrac{P_{ELC_{rated}} \times e_{ELC}}{HHV_{H_2}} & \text{if } P_{exc_{pvwt}}(t) \ge P_{ELC_{rated}} \times n_{ELC} \end{cases} \tag{12}$$

where, $P_{ELC_{rated}}$ represents the rated power of the electrolyzer, and n_{ELC}, denotes the number of electrolyzers. In this study, e_{ELC} is assumed to be 0.85.

3.5. Hydrogen tank

HT is used to store the hydrogen produced from excess energy through water electrolysis, as well as the hydrogen generated by the reformer when required.

To calculate the amount of hydrogen stored in the tank at time step t (1 h), the following Eq. (13) is used [6,45]:

$$E_{tank}(t) = E_{tank}(t-1) + P_{elc_tank}(t) - \frac{P_{tank_fc}(t)}{e_{tank}} \tag{13}$$

Here E_{tank} represents the existing capacity (kg) of the HT, P_{elc_tank} denotes the hydrogen amount (kg) transferred from the electrolyzer, P_{tank_fc} is the hydrogen amount (kg) supplied to the fuel cell, which means the hydrogen consumption of the fuel cell and e_{tank} indicates the efficiency of the HT, which is assumed to be 0.95.

3.6. Fuel cell

Fuel cells are energy conversion devices that produce electrical energy and heat through the electrochemical reaction of hydrogen with oxygen in the air. In the designed system, two types of hydrogen feed the fuel cell. The first is hydrogen generated from excess energy and converted in the electrolyzer, while the other is hydrogen produced directly by the reformer. The fuel cell's hydrogen consumption ($Cons_{H_2}$) during time interval t is determined using the following Eq. (14) [14,46]:

$$Cons_{H_2} = \frac{P_{tan\ k_{fc}}(t)}{e_{tank}} = \begin{cases} fcic \times P_{fc_r} \times \mathrm{i} + fcs \times P_{fc}(t) & if\ P_{def}(t) \le P_{fc_r} \times i \quad (i \le n) \\ fcic \times P_{fc_r} \times \mathrm{i} + fcs \times P_{fc_r} & if\ P_{def}(t) > P_{fc_r} \times \mathrm{i} \quad (i \le n) \end{cases} \tag{14}$$

where $fcic$ represents the fuel cell intercept coefficient (kg/hr/ kW_{rated}), fcs is the fuel cell slope coefficient (kg/hr/kW), P_{fc_r} stands for the rated power output of the fuel cell (kW), $P_{fc}(t)$ denotes the power output of the fuel cell for the time interval t (kW), P_{def} represents the deficient energy, n denotes the total number of fuel cells in the system, and i refers to the number of fuel cells used during the deficient energy state at time t. In this study, the values of $fcic$ and fcs are set to 0.08 kg/h/kW and 0.25 kg/h/kW, respectively [47].

In Eq. (14), i represents the total number of fuel cells required to meet the hydrogen consumption for compensating the energy deficit. If i is less than n (the total number of fuel cells in the population), only the necessary number of fuel cells will be activated to meet the demand at that time.

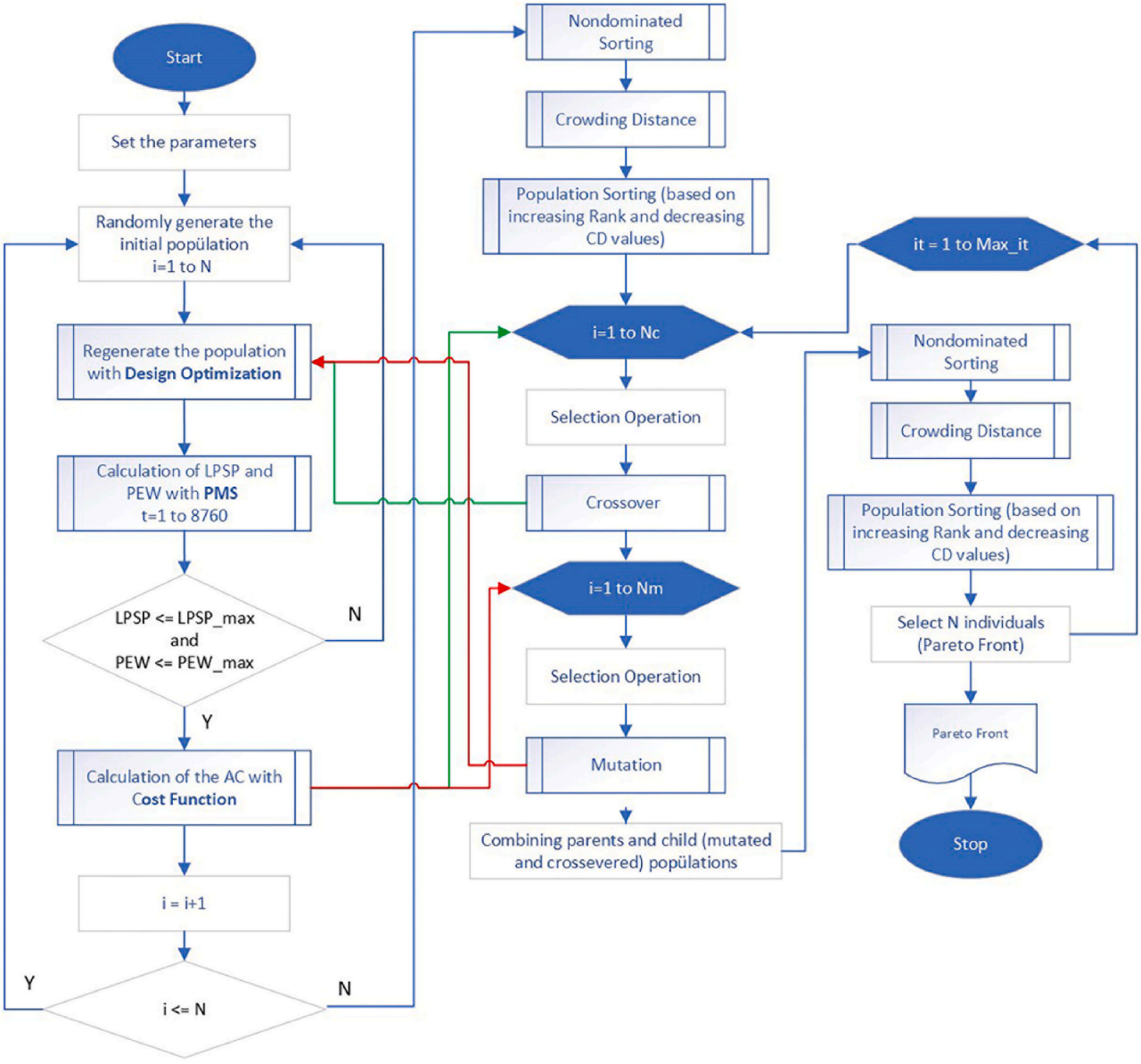


Fig. 4. Flowchart of i-NSGA-II.

The hydrogen consumption of the fuel cells is directly related to their rated power. Therefore, the algorithm optimizes the number of fuel cells by utilizing those with smaller rated power to meet the total required fuel cell capacity. In this way, the PMS optimizes the hydrogen consumption of the fuel cells, ensuring more efficient use of hydrogen.

Concerning the lower heating value of hydrogen ($LHV_{H_2} = 33.3\ kWh/kg$), the percentage of the fuel cell efficiency (η_{fc}) is calculated as follows [13]:

$$\eta_{fc}(\%) = \frac{P_{fc}}{LHV_{H_2} \times Cons_{H_2}} \times 100 \tag{15}$$

3.7. Reformer

Biogas is produced from the mixture of methane and carbon dioxide gases through the treatment of organic waste in an anaerobic digester (reactor) in an oxygen-free environment. In the designed hybrid system, the hydrogen obtained from the reformation of biogas is transferred directly from the reformer to the fuel cell and converted into electrical energy. According to the study in Ref. [38], the anaerobic reactor and the reformer are treated as a single system, with waste serving as the input and hydrogen as the output. The amount of waste required to be fed into the anaerobic reactor is constant. Based on the maximum hydrogen output of the reformer (P_{H_2}), it can be calculated as follows [38]:

$$W(kg) = \frac{P_{H_2}(kg)}{0,0454} \tag{16}$$

The energy produced by the fuel cell in hour t, using the hydrogen supplied from the reformer is calculated by rearranging Eq. (14) as follows:

$$P_{fc_rfr}(t) = \left(Cons_{H_2}(t) - fcic \times P_{fc_r}\right) / fcs \tag{17}$$

4. Multi-objective optimization

To address the multi-objective and multi-constrained optimization problem, i-NSGA-II is proposed. This algorithm is an enhancement of the system design optimization and advanced PMS algorithm, which are integrated with NSGA-II to effectively handle both the design and operational optimization of the HRES. The primary goal of this research is to determine the optimal configuration of HRES components to minimize ACS, LPSP, and PEW. The procedure and steps of i-NSGA-II are outlined in Fig. 4. The tri-objective optimization model can be expressed as the following Eq. (18) [9]:

$$\min F_{x,y} = \{ F_{AC}, F_{LPSP}, F_{PEW} \} \tag{18}$$

$$x = (i_{PV}, i_{WT}, i_{FC}\ i_{ELC}, i_{HT}, i_{RFR})$$

$$y = (\ n_{PV}, n_{WT},\ n_{FC}, n_{ELC}, n_{HT})$$

subject to

$$0 \le i_{PV}, i_{wt}, i_{FC}, i_{ELC}, i_{HT}, i_{RFR} \le S_{max}$$

$$1 \le n_{PV}, n_{WT}, n_{FC}, n_{ELC}, n_{HT} \le N_{max}$$

$$0 \le LPSP \le LPSP_{max}$$

$$0 \le PEW \le PEW_{max}$$

Here, $LPSP_{max}$ and PEW_{max} represent the maximum allowable values for LPSP and PEW, respectively, which are set at 5% to ensure that the system's energy waste and reliability losses remain within acceptable limits. Additionally, N_{max} denotes the maximum allowable number of each system component, while S_{max} specifies the maximum size value defined for each component.

Table 3
Sizing options for system components.

System Component	Size Options	Number Options
PV	0 kW, 1 kW, 2 kW, 3 kW, 4 kW, 5 kW, 6 kW, 7 kW, 8 kW, 9 kW, 10 kW	1 to 15
WT	0 kW, 1 kW, 2 kW, 3 kW, 4 kW, 5 kW, 6 kW, 7 kW, 8 kW, 9 kW, 10.5 kW	1 to 15
Fuel cell	0 kW, 1 kW, 2 kW, 3 kW, 4 kW, 5 kW, 6 kW	1 to 15
HT	0 kg, 1 kg, 1.5 kg, 2 kg, 2.5 kg, 3 kg, 3.5 kg, 4 kg	1 to 15
Electrolyzer	0 kW, 1 kW, 2 kW, 3 kW, 4 kW, 5 kW	1 to 15
Battery	0 kW, 1 kW, 2 kW, 3 kW, 4 kW, 5 kW	1 to 20
Reformer	0 kg, 0.1 kg, 0.2 kg, 0. 3 kg, 0.4 kg, 0.5 kg, 0.6 kg, 0.8 kg	–

In this study, the decision variables considered within the multi-objective optimization framework are structured in two distinct forms, based on size and number. Additionally, the size of a system component encompasses not only its nominal output power but also characteristics specific to its type, such as the temperature coefficient of the PV, the efficiency coefficient of the fuel cell, and the cut-in and cut-off speeds of the WT, among others.

Each system component is evaluated based on its size (type) and number. The number of each component is randomly generated by the algorithm within a specified range, as shown in Table 3, to explore a wide spectrum of possible configurations. This diverse range of size and number options ensures that the optimization process explores a comprehensive set of configurations to identify solutions that balance cost, reliability, and system efficiency.

Based on the data in Table 3, the number of configuration possibilities for this system is 246,985,200,000 for Scenario 1 and 329,313,600,000 for Scenario 2. Given the vast number of possible configurations, i-NSGA-II is employed to scan this large solution space and achieve Pareto optimal solutions within acceptable timeframes.

Table 4 summarizes the key parameters that define the configuration of i-NSGA-II. These settings guide the optimization process in exploring and identifying the optimal configurations for the HRES, balancing the search for more Pareto optimal solutions with minimizing optimization time.

In this study, MOPSO is employed to validate and compare the performance of the developed i-NSGA-II. PSO, an evolutionary algorithm, draws inspiration from the behaviours observed in flocks of birds, fish, and insects [48]. The MOPSO, an extension of the original PSO, is capable of finding Pareto front solutions for multiple conflicting objectives, similar to algorithms such as NSGA-II. For an in-depth understanding of the MOPSO algorithm's operation, refer to Refs. [49,50].

Table 5 provides the parameters of MOPSO. To ensure a fair and consistent comparison, identical constraints were applied in both algorithms. Additionally, assumptions such as treating the anaerobic reactor and reformer as a single system, alongside consistent parameters like the technical specifications of system components and meteorological data, were maintained throughout the analysis. This approach focuses the evaluation solely on the differences in algorithmic performance.

4.1. Objective functions

Table 6 summarizes the objectives, corresponding equations, and abbreviations used in the i-NSGA-II. The objectives include economic performance, reliability, and system efficiency, each quantified through

Table 4
Parameters of i-NSGA-II.

Parameters	Values	Parameters	Values
Population size	50	Mutation rate	0.2
Maximum generation	100	Mutation probability	0.02
Crossover rate	0.7		

Table 5
Parameters used in MOPSO.

Parameters	Values	Parameters	Values
Maximum number of Iterations	100	Global learning coefficient	2
Population size	100	Number of grids per dimension	7
Repository size	50	Inflation rate	0.1
Inertia weight	0.5	Leader selection pressure	2
Inertia weight damping rate	0.99	Deletion selection pressure	2
Personal learning coefficient	1	Mutation rate	0.1

specific mathematical formulations. In the ACS formulation, the nominal interest rate ($ir_{nominal}$) is taken as 0.04 and the annual inflation rate (fr) is 0.03.

In Table 6, the useful life of the project (project lifetime), denoted by R, is assumed to be 25 years. Due to space constraints, it is not feasible to present all the parameters related to the types of system elements and the model assumptions in a table. However, the key economic parameters are summarized in Table 7.

4.2. Decision variables

The i-NSGA-II defines the following thirteen decision variables for this study, as represented in Table 8.

4.3. System design optimization

The proposed system design optimization approach differs from the size optimization approach found in the literature because it determines whether a component will be included in the system and, if included what its type (size) and quantity will be. Therefore, the existence of an element is denoted by two distinct structures, each representing its size and number, with the total size being the product of these two. Additionally, this approach includes the necessary steps to design a system where hydrogen can be present not only as an energy backup but also as a direct energy source.

In Scenario 1, to ensure uninterrupted hydrogen production, if the system includes a fuel cell, it is arranged to have a reformer of a size that can meet the minimum hydrogen consumption and not exceed the maximum hydrogen consumption of the fuel cell. Furthermore, to store excess hydrogen, a HT is ensured to be included in the system alongside the fuel cell in this case. Similarly, if an electrolyzer is present in the randomly generated initial population, it is arranged to include the fuel cell-HT pair it will work with. The system is designed to ensure the integrity of fuel cell-electrolyzer-HT or Reformer-fuel cell-HT energy systems in a randomly generated initial population.

In Scenario 2, instead of an electrolyzer, a battery is included in the system to store excess energy from PVs and WTs. In this case, the system includes both a battery and hydrogen storage, ensuring energy balance.

Table 8
Decision variables.

System Element Type (Size)	Abbreviation	System Element Number	Abbreviation
PV type (kW)	i_{PV}	PV number	n_{PV}
WT type (kW)	i_{WT}	WT number	n_{WT}
Fuel cell type (kW)	i_{FC}	Fuel cell number	n_{FC}
Electrolyzer type (kW)	i_{ELC}	Electrolyzer number	n_{ELC}
Battery type (kW)	i_{BAT}	Battery number	n_{BAT}
HT type (kg)	i_{HT}	HT number	n_{HT}
Reformer type (kg)	i_{RFR}	–	

Table 6
Objective functions applied in i-NSGA-II.

Objectives	Equations	Abbreviations
The economic objective [51]	$ACS\left(\frac{\$}{year}\right) = \sum_{i=1}^{n} N_i \times \left[\left(CC_i + RC_i \times F_i\left(ir, L_i, y_i\right)\right) \times CRF(ir, R) + OMC_i\right]$ $ir = \frac{ir_{nominal} - fr}{1 + fr}$ $CRF\left[ir, R\right] = \frac{ir\left[1 + ir\right]^R}{\left[1 + ir\right]^R - 1}$ $F = \sum_{n=1}^{Y_i} \frac{1}{\left[1 + ir\right]^{L_i \times n}}$ *If R is divisible by L;* $Y = \frac{R}{L} - 1$ *If R is not divisible by L;* $Y = \frac{R}{L}$	n: The number of system elements N: the number of component *i* CC_i: The capital cost ($/kW) of component i OMC_i: The annual operation and maintenance cost *ir:* real interest rate $ir_{nominal}$: Nominal interest rate *fr*: annual inflation rate *CRF*: The capital recovery factor *R:* useful life of the project L_i: The useful life of component i Y_i: the number of replacements of component i
The reliability objective	$LPSP\ (\%) = \frac{\sum_{t=1}^{8760} LPS(t)}{\sum_{t=1}^{8760} P_{load}(t)}$ $LPS(t) = P_{load}(t)/e_{conv} - P_{sist}(t) - P_fc_def(t) - P_{bat}(t)$	$P_{load}(t)$: The hourly load demand $LPS(t)$: The hourly unmet load $P_{sist}(t)$: The hourly total power produced by the hybrid system $P_{bat}(t)$: The battery's power at time t e_{conv}: The converter efficiency
The system efficiency objective	**For Scenario 1:** $PEW = \frac{\sum_{t=1}^{8760} P_{exc_{pvwt}}(t) - P_{ren_elc}(t)\right]}{\sum_{t=1}^{8760} P_{syst}(t)}$ **For Scenario 2:** $PEW = \frac{\sum_{t=1}^{8760} P_{exc_{pvwt}}(t) - P_{ren_bat}(t)\right]}{\sum_{t=1}^{8760} P_{syst}(t)}$ $P_{syst}(t) = P_{pv}(t) + P_{wt}(t) + P_{rfr}(t)$	$P_{ren_elc}(t)$: The excess power transferred from PVs and WTs to electrolyzers $P_{ren_bat}(t)$: The excess power transferred from PVs and WTs to batteries

Table 7
Cost values and lifetimes of system components [9,19,51–53],[a].

	PV	WT	Battery	Fuel Cell	Electrolyzer	HT	Reformer
Acquisition cost	700 ($/kW)	3400 ($/kW)	375 ($/kW)	3000 ($/kW)	1500 ($/kW)	1000 ($/kg)	3200 ($/0,1 kg)
Operation and maintenance cost ($/year)	14	68	11	88	20	50	225
Replacement cost ($)	–	–	375 ($/kW)	3000 ($/kW)	1000 ($/kW)	–	–
Lifetime (year)	25	25	10	5	10	25	25

[a] Cost data were obtained from these sources, and approximate/average values were used.

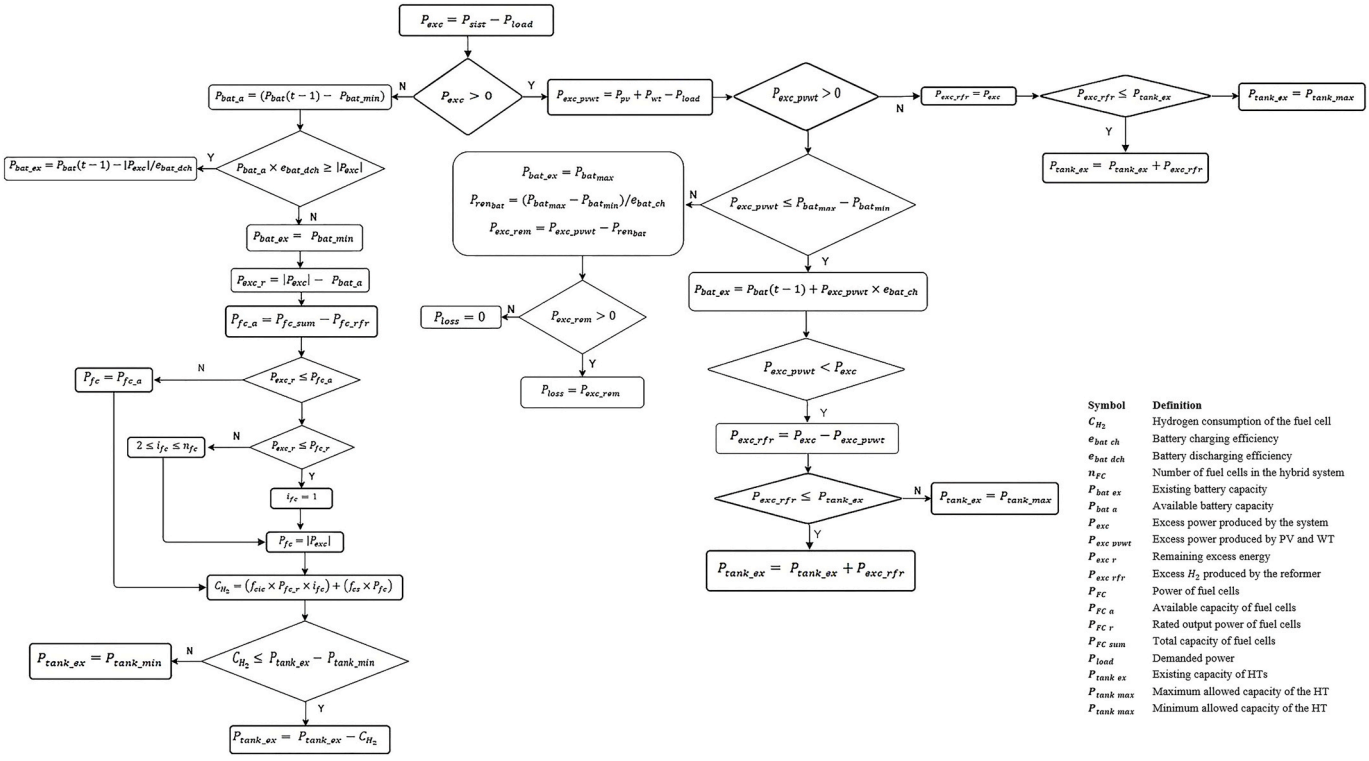


Fig. 5. A brief diagram of developed PMS.

When a reformer is present, a fuel cell is also included to produce electricity from the hydrogen, and a HT is included to store excess hydrogen. Additionally, as in Scenario 1, the reformer size is adjusted based on the minimum and maximum hydrogen consumption of the fuel cell. The system ensures that the energy generated by the renewable sources is efficiently stored in either the battery or HT, optimizing the energy supply.

In summary, for both scenarios, the initial population is generated randomly, ensuring that the system includes at least two renewable energy sources and a balanced energy storage solution.

An important point here is that the type (size) of a system component not only refers to its nominal output power but also comprises various characteristics such as the installation height and cut-in/cut-out speed of the WTs; surface roughness length and temperature coefficient of the PV, and hydrogen consumption of the fuel cell, etc. These values are included in the calculations and are read from the relevant external table for each component within the PMS.

4.4. Power management strategy (PMS)

The PMS regulates energy exchanges among the hybrid system components to optimize system configuration by minimizing ACS, LPSP, and PEW. This study utilizes hydrogen both as an uninterrupted power source and for excess energy storage. The developed PMS ensures effective energy flow between renewable energy sources and improves both system reliability and efficiency by employing various energy storage options tailored to each scenario.

In Scenario 1, the system utilizes hydrogen storage as the primary solution for managing energy flow. Unlike conventional hydrogen storage systems in the literature, this scenario incorporates an enhanced PMS featuring hydrogen sourced from two different pathways. Excess electricity generated by PVs and WTs is directed to the electrolyzer, where it is converted into hydrogen. Additionally, when PV and WTs generation meets or exceeds the load demand, surplus hydrogen produced by the reformer is also stored in HTs. During periods of insufficient renewable energy production, the stored hydrogen is utilized by fuel cells to supply power, ensuring an uninterrupted energy flow. This dual utilization of the electrolyzer and reformer for hydrogen production, combined with efficient storage in HTs, significantly improves the system's reliability and operational flexibility.

With Scenario 2, a hybrid storage system has been aimed at achieving the same objectives, where excess hydrogen from only the reformer is stored in HTs and batteries are used for excess energy from PVs and WTs. This hybrid approach aims to diversify storage options, thereby improving the system's overall flexibility and efficiency. Batteries handle short-term fluctuations in renewable energy production, whereas hydrogen storage addresses long-term energy needs. Fig. 5 presents a flowchart summarizing the working principle of the PMS for Scenario 2, which includes hydrogen and battery storage. For Scenario 1 PMS, detailed explanations and a summary flow diagram can be examined from the author's previous work [9].

5. Results and discussions

5.1. Multi-objective optimization results for three selected locations in Türkiye

A three-objective optimization algorithm was designed in MATLAB to minimize ACS, LPSP, and PEW values. The i-NSGA-II was developed in conjunction with the system design optimization and an advanced PMS and, aimed at optimizing hydrogen utilization to improve system reliability and efficiency. The Pareto solutions obtained using i-NSGA-II

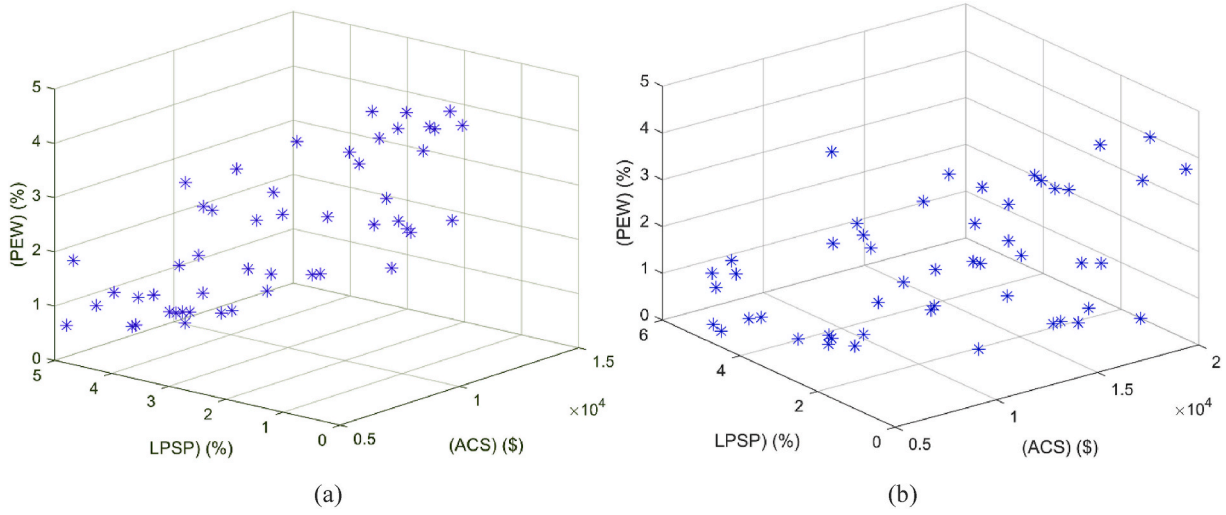


Fig. 6. The Pareto front for Çanakkale **(a)** Scenario 1 **(b)** Scenario 2.

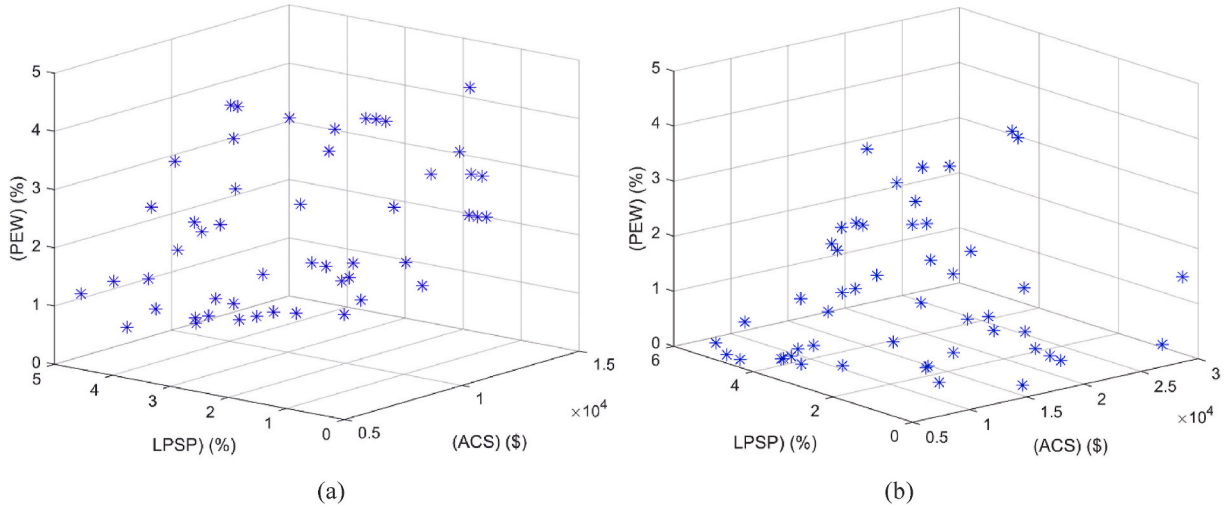


Fig. 7. The Pareto front for Çeşme **(a)** Scenario 1 **(b)** Scenario 2.

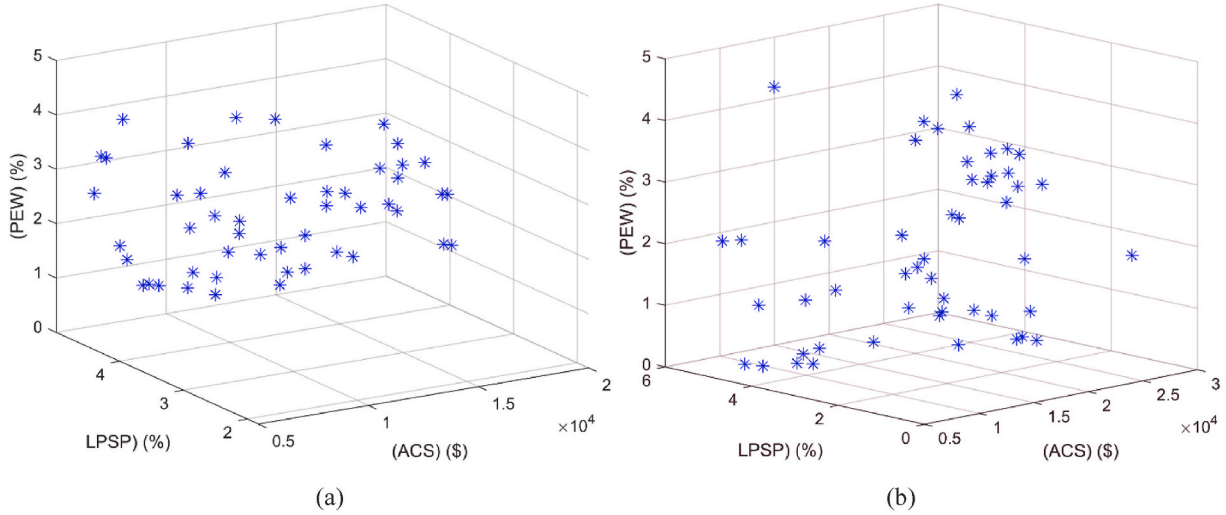


Fig. 8. The Pareto front for Kahramanmaraş **(a)** Scenario 1 **(b)** Scenario 2.

Table 9
Selected solutions of the Pareto front for Çanakkale according to scenario 1.

		i_{PV} (kW)	n_{PV}	i_{WT} (kW)	n_{WT}	i_{FC} (kW)	n_{FC}	i_{ELC} (kW)	n_{ELC}	i_{HT} (kg)	n_{HT}	i_{RFR} (kg)	ACS) ($/y)	LPSP (%)	PEW (%)
1	Lowest ACS Solution	4	1	1	3	3	1	0	–	1,5	3	0.8	5129.25	4.72	1.89
2	Lowest LPSP Solution	3	1	1	6	3	2	1	5	4	15	0.8	11435.06	0.55	4.47
3	Lowest PEW Solution	3	1	4	1	3	2	3	1	4	15	0.8	10586.14	4.97	0.10

Table 10
Selected solutions of the Pareto front for Çanakkale according to scenario 2.

		i_{PV} (kW)	n_{PV}	i_{WT} (kW)	n_{WT}	i_{FC} (kW)	n_{FC}	i_{BAT} (kW)	n_{BAT}	i_{HT} (kg)	n_{HT}	i_{RFR} (kg)	ACS ($/y)	LPSP (%)	PEW (%)
1	Lowest ACS Solution	2	2	1	3	3	1	1	1	4	1	0.8	5031.17	4.73	1.48
2	Lowest LPSP Solution	1	3	1	7	3	2	5	17	4	15	0.8	15417.91	0.15	4.76
3	Lowest PEW Solution	1	2	1	4	3	1	5	8	4	4	0.8	7945.56	4.03	0.00

Table 11
Solutions of the Pareto front for Çeşme according to scenario 1.

		i_{PV} (kW)	n_{PV}	i_{WT} (kW)	n_{WT}	i_{FC} (kW)	n_{FC}	i_{ELC} (kW)	n_{ELC}	i_{HT} (kg)	n_{HT}	i_{RFR} (kg)	ACS ($/y)	LPSP (%)	PEW (%)
1	Lowest ACS Solution	3	1	1	3	3	1	1	1	1.5	2	0.8	5128.30	4.60	1.27
2	Lowest LPSP Solution	0	–	1	7	3	2	1	5	4	15	0.8	11533.57	0.27	3.37
3	Lowest PEW Solution	0	–	1	4	3	2	2	2	4	15	0.8	10696.06	4.86	0.06

for Scenario 1 and Scenario 2 in Çanakkale, Çeşme, and Kahramanmaraş provinces are presented in Fig. 6, Fig. 7, and Fig. 8, respectively.

Since the Pareto front solutions are non-dominant with respect to each other, selecting a single optimal solution is not feasible. This is an inherent characteristic of the multi-objective optimization approach, which simultaneously optimizes three conflicting objectives. Consequently, solutions with the minimum values for each objective function (ACS, LPSP, and PEW) have been selected from the Pareto solutions and are presented in detail in Tables 9–14 for three locations: Çanakkale, Çeşme, and Kahramanmaraş.

Tables 9 and 10 present the objective function values and system configurations for the minimum ACS, LPSP, and PEW solutions selected from the Pareto fronts for Scenario 1 and Scenario 2 for Çanakkale. The following evaluation summarizes the key differences between these two scenarios based on these parameters. Scenario 2 shows a 2% improvement in ACS, no change in LPSP (0%), and a 22% reduction in PEW compared to Scenario 1 for the lowest ACS solution. When comparing the lowest LPSP solutions, Scenario 2 achieves a 73% improvement in LPSP, along with a 35% increase in ACS and a 6% rise in PEW. Finally, for the lowest PEW solution, Scenario 2 outperforms Scenario 1 with a 25% reduction in ACS, a 19% decrease in LPSP, and a complete elimination of PEW. These results highlight that Scenario 2 significantly enhances system reliability and efficiency, while maintaining competitive performance in terms of ACS with continuous hydrogen production and

Table 12
Solutions of the Pareto front for Çeşme according to scenario 2.

		i_{PV} (kW)	n_{PV}	i_{WT} (kW)	n_{WT}	i_{FC} (kW)	n_{FC}	i_{BAT} (kW)	n_{BAT}	i_{HT} (kg)	n_{HT}	i_{RFR} (kg)	ACS ($/y)	LPSP (%)	PEW (%)
1	Lowest ACS Solution	2	2	1	2	3	1	3	2	1	3	0.8	5151.63	4.99	0.29
2	Lowest LPSP Solution	0	–	1	8	3	2	5	20	3	7	0.8	14346.29	0.04	4.73
3	Lowest PEW Solution	2	1	2	2	3	1	3	6	4	13	0.8	8707.64	3.86	0.00

Table 13
Solutions of the Pareto front for Kahramanmaraş according to scenario 1.

		i_{PV} (kW)	n_{PV}	i_{WT} (kW)	n_{WT}	i_{FC} (kW)	n_{FC}	i_{ELC} (kW)	n_{ELC}	i_{HT} (kg)	n_{HT}	i_{RFR} (kg)	ACS ($/y)	LPSP (%)	PEW (%)
1	Lowest ACS Solution	2	2	1	5	3	1	0	–	3.5	2	0.8	5614.62	4.51	3.46
2	Lowest LPSP Solution	0	–	2	7	3	2	2	5	4	15	0.8	13589.69	1.82	3.64
3	Lowest PEW Solution	2	1	1	5	3	2	2	2	4	15	0.8	11000.09	4.57	0.50

Table 14
Solutions of the Pareto front for Kahramanmaraş according to scenario 2.

		i_{PV} (kW)	n_{PV}	i_{WT} (kW)	n_{WT}	i_{FC} (kW)	n_{FC}	i_{BAT} (kW)	n_{BAT}	i_{HT} (kg)	n_{HT}	i_{RFR} (kg)	ACS ($/y)	LPSP (%)	PEW (%)
1	Lowest ACS Solution	3	1	1	5	3	1	0	–	4	3	0.8	5821.96	4.88	2.19
2	Lowest LPSP Solution	0	–	1	12	3	2	5	19	4	15	0.8	16928.69	0.30	3.43
3	Lowest PEW Solution	0	–	1	8	3	1	4	20	4	12	0.8	12789.41	4.55	0.00

hybrid storage.

In Tables 11 and 12, the objective function values and system configurations for the minimum ACS, LPSP, and PEW solutions from the Pareto fronts are provided for both Scenario 1 and Scenario 2 in Çeşme. For the lowest ACS solution, Scenario 2 shows no change in ACS (0%), an 8% increase in LPSP, and a 77% reduction in PEW compared to Scenario 1. In the lowest LPSP solutions, Scenario 2 demonstrates an 85% improvement in LPSP, accompanied by a 24% increase in ACS and a 40% rise in PEW. For the lowest PEW solution, Scenario 2 outperforms Scenario 1 with a 19% reduction in ACS, a 21% decrease in LPSP, and a complete elimination of PEW. These results indicate that Scenario 2 is particularly effective in reducing PEW and improving reliability, offering significant potential for hybrid storage systems in Çeşme. Additionally, when prioritizing reliability and cost minimization, hydrogen storage alone (Scenario 1) presents a viable alternative for Çeşme.

When comparing the solutions for Kahramanmaraş in Tables 13 and 14, Scenario 2 results in a 4% and 8% increase in ACS, and LPSP, respectively with a 37% reduction in PEW for the lowest ACS solution, relative to Scenario 1. For the lowest LPSP solution, Scenario 2 achieves 84% and a 6% reduction in LPSP and PEW, despite a 25% increase in ACS. In the lowest PEW solution, Scenario 2 eliminates PEW, results in a slight decrease in LPSP (0.44%), but results in a 16% increase in ACS compared to Scenario 1.

It is observed that the cost of achieving a more reliable and efficient HRES at the selected location in Kahramanmaraş is higher compared to the other locations. This is due to the relatively lower average wind speed and significantly lower temperature average compared to the other two locations, resulting in a lower renewable energy production potential.

Additionally, it is observed that PV are absent in the minimum LPSP solutions for both Çeşme and Kahramanmaraş. Despite the high solar radiation potential in both cities (see Table 1), PVs are not included in the solutions. This is because, although PVs generate excess energy in the summer, they fail to provide sufficient production in the winter, making it difficult to maintain the balance between energy supply and demand throughout the year. These results also demonstrate how significantly regional conditions influence the configuration of HRES.

In the following sections, as highlighted by the simulation results, solutions featuring acceptable and balanced values for all three objectives consistently include both PV and WTs, which complement each other effectively. Another noteworthy aspect of the optimal solutions is the uniform setting of the reformer capacity at 0.8 kg. This maximum limit for reformer capacity is prioritized by the PMS to ensure a consistent and uninterrupted renewable energy supply, thereby enhancing system reliability and efficiency.

5.2. Simulation results

Achieving a well-balanced Pareto front in the multi-objective optimization of hybrid energy systems is only possible with a well-designed PMS. To demonstrate the energy flow between system components and the effectiveness of storage systems in both scenarios, annual simulations on an hourly basis were conducted using the developed PMS. To validate the mathematical models of the system components, system design optimization, and the PMS, intermediate solutions for each of the three cities were selected from the Pareto optimal solutions and thoroughly analyzed. By evaluating days from three different seasons in these cities with diverse climate characteristics, the behaviour of the PMS under varying conditions has been demonstrated. The selected solutions for Scenarios 1 and 2 among the Pareto solutions for Çanakkale, Çeşme, and Kahramanmaraş can be found in Table 15.

When considering the Pareto optimal solutions in Tables 15 and in Scenario 2, the use of batteries has generally provided PEW and better LPSP values compared to electrolyzers. Çeşme Scenario 2 stands out with the lowest cost and energy waste, while Çanakkale Scenarios 1 and 2 provide balanced solutions in terms of cost and efficiency. However, while battery use has increased costs in Kahramanmaraş, it reduces energy waste and improves supply security. The choice between batteries and electrolyzers varies depending on regional energy potential and cost-effectiveness.

In addition, as observed in the minimum solutions for all three objectives in the previous section, for average solutions in Table 15, a continuous and stable hydrogen-producing reformer has been preferred as the main power source, followed by WTs generating electricity from variable wind energy throughout the year. Conversely, PVs have been integrated into the system to a lesser extent due to their effectiveness limited to daylight hours and predominantly during summer months.

5.2.1. Simulation results for Çanakkale

As detailed in Section 2.2, Çanakkale has the highest wind potential among the three selected cities. As seen in Figs. 9 and 11 in the annual simulation based on monthly averages, WTs have the highest share after the reformer. In Fig. 9, during the summer months, due to high solar radiation, the energy production from PVs increases, whereas that from WTs decreases. The dark green bars (P_rfr) represent the energy generated by the fuel cell using hydrogen produced by the reformer, while the light green bars (P_fc_def) indicate the energy produced by the fuel cell using hydrogen from the electrolyzer in the event of an energy deficit. Particularly during the winter months, the fuel cell not only generates electricity from the reformer's hydrogen but also compensates for the energy shortfall with its remaining capacity. The red diamonds (E_dump) illustrate the wasted energy, and the yellow stars (P_loss) denote the unmet load. It can be observed that wasted energy occurs especially in February, March, and October, when the total energy production of the system exceeds the load demand.

Fig. 10 shows the energy flow for Çanakkale in Scenario 1 on an autumn day. Due to shorter daylight hours, solar energy production (P_pv) is low, while WTs power (P_wt) plays a larger role. When renewable energy generation is insufficient, the fuel cell compensates for the shortfall (P_fc_def), and any unmet load is represented by the white triangles (P_loss = 0). The yellow squares (P_rfr_tank) indicate the surplus hydrogen produced by the reformer (P_rfr, shown by the orange bars) and stored in the HT. Fig. 10 emphasizes the critical role of continuous hydrogen production from the reformer in maintaining energy balance during periods of insufficient wind and solar energy, such as this autumn day.

Fig. 11 shows the energy flow for Çanakkale in Scenario 2, where the energy flow is balanced through the PMS. In this scenario, the excess

Table 15
Intermediate Solutions selected from the Pareto front.

		i_{PV} (kW)	n_{PV}	i_{WT} (kW)	n_{WT}	i_{FC} (kW)	n_{FC}	i_{elc}/i_{bat} (kW)	n_{ELC}/n_{bat}	i_{HT} (kg)	n_{HT}	i_{RFR} (kg)	*ACS* *($/y)*	*LPSP* *(%)*	*PEW* *(%)*
Çanakkale	Autumn day Scenario 1	3	1	4	2	3	2	1	5	4	15	0.8	11701.65	1.56	2.22
	Autumn day Scenario 2	1	3	2	3	3	1	5	11	1.5	15	0.8	10108.54	1.66	1.36
Çeşme	Winter day Scenario 1	1	1	1	5	3	2	1	2	3	10	0.8	9262.86	1.57	1.89
	Winter day Scenario 2	2	1	1	5	2	2	5	4	4	5	0.8	7938.59	1.33	1.02
Kahramanmaraş	Summer day Scenario 1	3	1	2	4	3	2	3	5	4	6	0.8	11435.52	2.98	2.96
	Summer day Scenario 2	3	1	1	8	3	1	5	20	4	15	0.8	14547.52	1.61	2.76

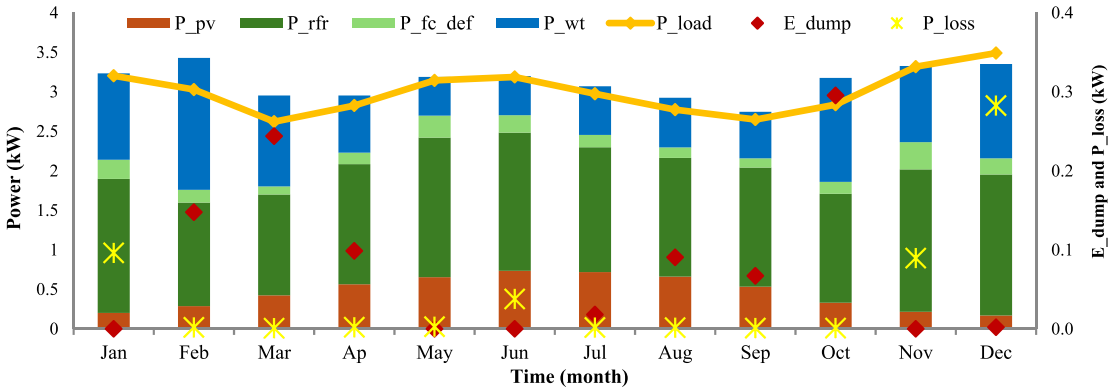


Fig. 9. Annually energy flow, wasted energy and loss of load for Çanakkale according to Scenario 1.

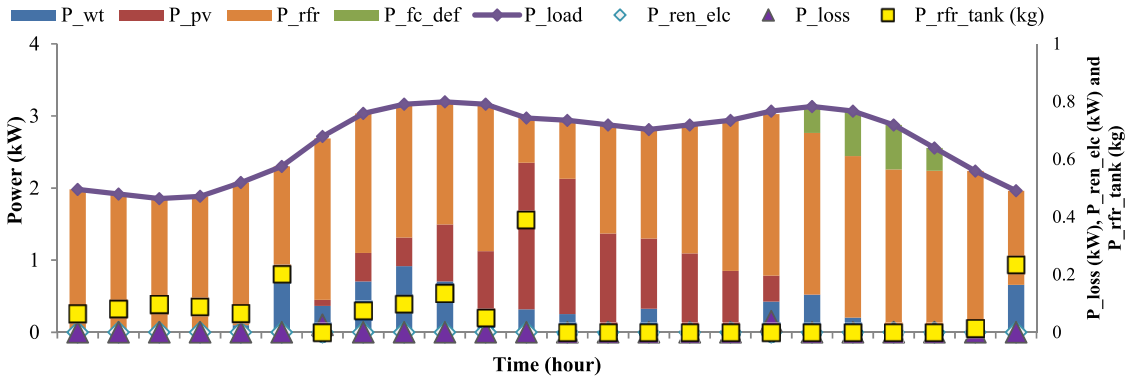


Fig. 10. Energy flow for Çanakkale according to Scenario 1 of an autumn day.

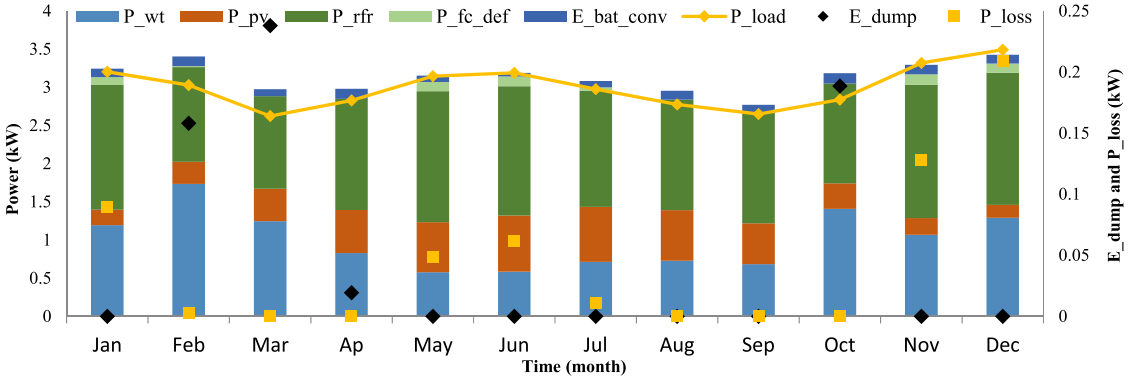


Fig. 11. Annually Energy flow, Wasted energy and loss of load for Çanakkale according to Scenario 2.

energy generated by the PVs and WTs is stored in the battery. During energy deficits, the energy produced by the battery (E_bat_conv) is used alongside the energy generated by the fuel cell (P_fc_def) to meet the load demand. This scenario highlights the role of battery storage in complementing the hydrogen storage system, ensuring a stable energy supply even during periods of low renewable generation.

Fig. 12 illustrates the energy flow on an autumn day in Scenario 2, similar to Fig. 10. When PV and WT are insufficient to meet the load demand, the reformer plays a critical role in maintaining energy balance. In this scenario, excess energy generated by PV and WT is directed to the battery (P_ren_bat). During the evening hours (19:00–22:00), when renewable sources are inadequate and there is insufficient hydrogen in HT, battery (P_bat_conv) is activated to compensate for the energy deficit. This energy flow demonstrates the combined contribution of renewable sources (PV, WT, and the reformer), with the battery ensuring energy supply continuity when the HT cannot meet the deficit energy.

5.2.2. *Simulation results for Çeşme*

Fig. 13 illustrates the monthly averages of the annual energy flow for Çeşme under Scenario 1. It highlights the significant contribution of WTs power (P_wt) alongside the reformer (P_rfr) throughout the year. The energy deficit is compensated by the fuel cell using hydrogen stored in the tank, represented by the light green bars (P_fc_def). The light blue diamond indicate dumped energy (E_dump), which occurs when the HT cannot store the excess energy. In cases where the HT is insufficient to compensate for the energy deficit with the fuel cell (P_fc_def), the unmet load (P_loss) is represented by the yellow stars.

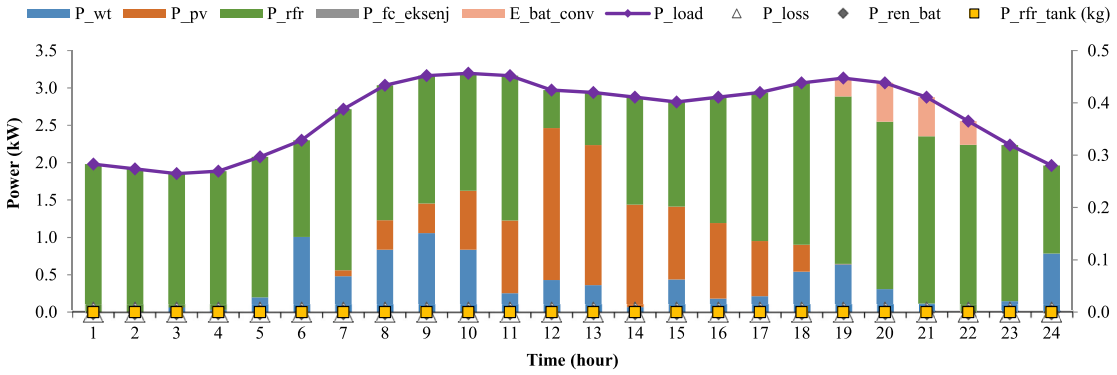


Fig. 12. Energy flow for Çanakkale according to Scenario 2 of an autumn day.

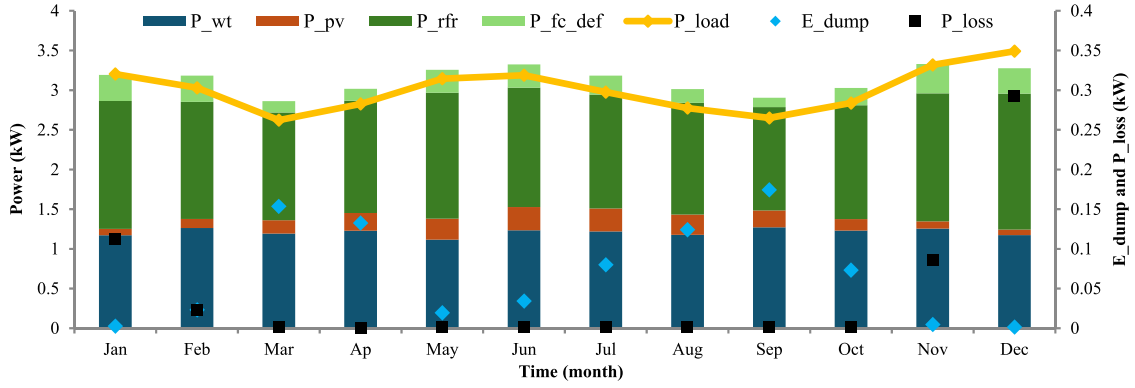


Fig. 13. Annually energy flow, wasted energy and loss of load for Çeşme according to Scenario 1.

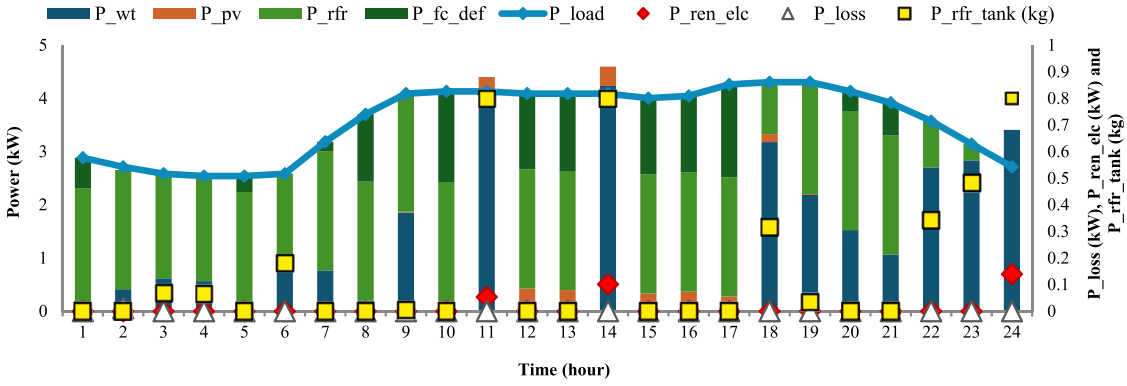


Fig. 14. Energy flow for Çeşme according to Scenario 1 of a winter day.

Fig. 14 shows the distribution and flow of energy sources over 24 h in Çeşme on a winter day according to Scenario 1. WTs power (P_wt) serves as a substantial and consistent energy source throughout the day, while PVs power (P_pv) peaks around noon and diminishes towards the evening. During hours of ample wind power in the evening, hydrogen generated by the reformer is transferred to storage tanks (P_rfr_tank). Conversely, when both wind and solar energies are insufficient or absent, fuel cells activate to compensate for the energy shortfall (P_fc_def). At 11, 14, and 24 o'clock, it is observed that the energy produced by PVs and WTs exceeds the demand, and the surplus energy is transferred to the electrolyzer to be converted into hydrogen for storage in the tank (P_ren_elc).

Fig. 15 illustrates the monthly averages of the annual energy flow for Çeşme under Scenario 2. It shows the seasonal variations in PVs power generation, which increases in summer and decreases in winter. The energy shortfall during the winter months is compensated by the fuel cell and battery, ensuring the system operates efficiently during the summer. The dark blue and light green bars represent E_bat_conv and P_fc_def, respectively, demonstrating how the hybrid storage system balances the energy flow. This minimizes the unmet load (P_loss) and dumped energy (E_dump) amounts.

Fig. 16 presents simulations for a winter day in Çeşme under Scenario 2. As solar energy decreases, WTs provide a significant portion of the energy demand throughout the day. When total renewable power is sufficient to meet the entire load demand, hydrogen produced by the reformer is stored in the tank (P_rfr_tank). At 11:00 and 14:00, the total energy produced by PV and WTs exceeds the load demand, and all the hydrogen produced by the reformer is sent to the tank. The excess energy from PV and WTs (P_ren_bat) is also sent to the battery. When PV, wind, and the reformer are insufficient to meet the load demand, the fuel

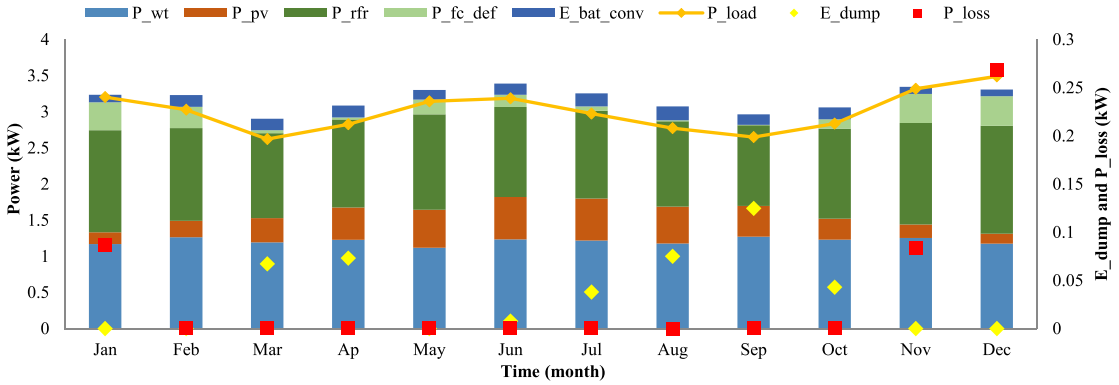


Fig. 15. Annually energy flow, wasted energy and loss of load for Çeşme according to Scenario 2.

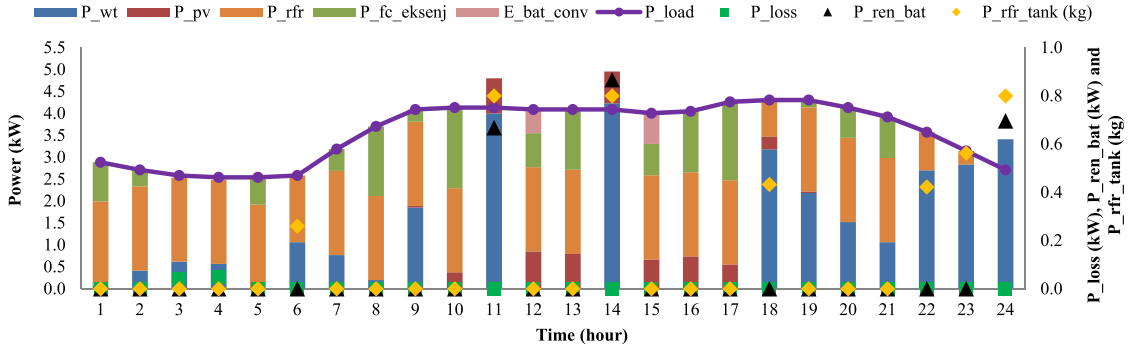


Fig. 16. Energy flow for Çeşme according to Scenario 2 of a winter day.

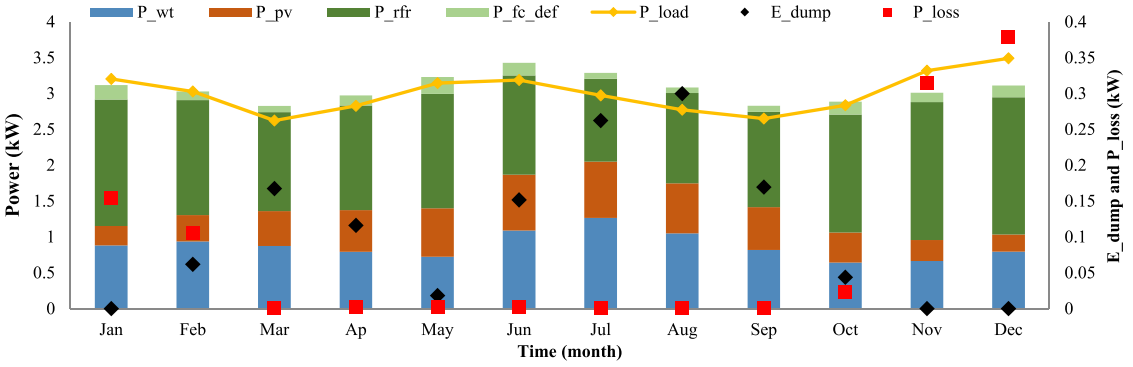


Fig. 17. Annually energy flow, wasted energy and loss of load for Kahramanmaraş according to Scenario 1.

cell (P_fc_def) and battery (P_bat_conv) are activated to cover the energy deficit.

As a result of the simulations, it is seen that the solar energy potential in Çeşme is very low in winter and wind energy continues throughout the year and at almost all hours of the day. This explains why WTs are preferred instead of PVs in Tables 11 and 12 2nd and 3rd solutions to ensure system reliability and to create a more efficient system by preventing energy waste.

5.2.3. *Simulation results for Kahramanmaraş*

According to the simulation results for Kahramanmaraş in Scenario 1, as observed in Fig. 17, during the summer months, the increased share of solar and wind energy is sufficient to meet the energy demand. However, in the winter months, the decreased solar potential combined with a drop in wind energy results in an energy deficit (P_loss).

Fig. 18 presents a graph summarizing the simulation results for a summer day in Kahramanmaraş according to Scenario 1. Solar power significantly contributes to the energy supply from around 8:00 to 17:00, peaking at midday. Wind power provides a contribution in the second half of the day. During periods of excess energy at midday, the produced excess hydrogen from the reformer is stored in the tank (P_rfr_tank). In the evening hours, when solar power is not available and wind power decreases, the energy deficit is met by the fuel cell (P_fc_def), which generates energy using the hydrogen stored in the tanks.

Fig. 19 illustrates the monthly averages of the annual energy flow, wasted energy, and loss of load for Kahramanmaraş under Scenario 2, highlighting the seasonal variations in energy generation and the efficiency of the hybrid storage system.

In hourly simulations conducted for a summer day in Kahramanmaraş according to Scenario 2, as seen in Fig. 20, it is observed that the energy produced by PVs and WTs exceeds the load

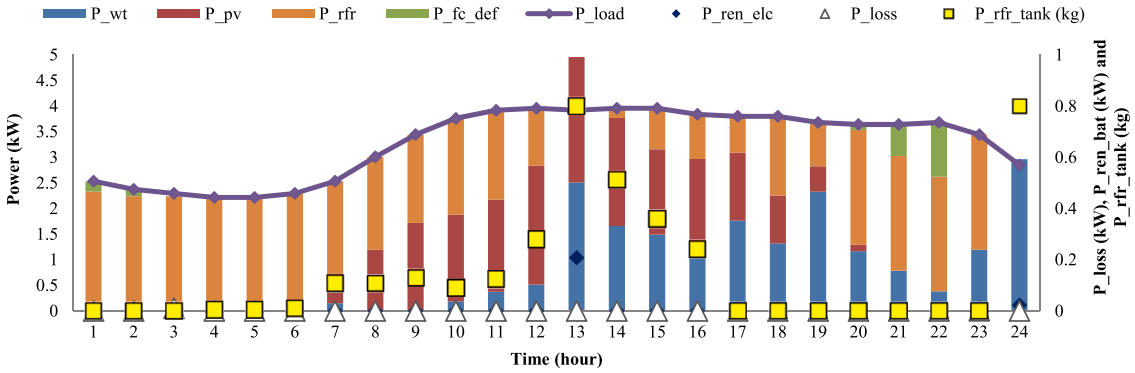


Fig. 18. Energy flow for Kahramanmaraş according to Scenario 1 of a summer day.

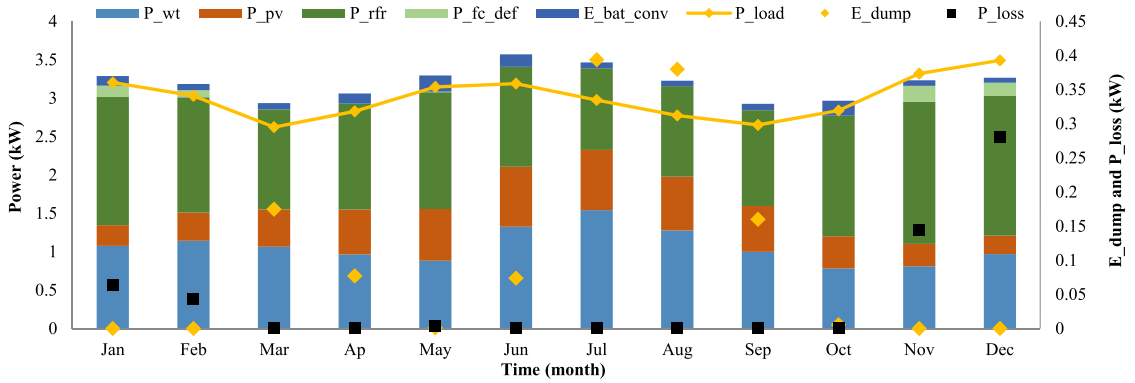


Fig. 19. Annually energy flow, wasted energy and loss of load for Kahramanmaraş according to Scenario 2.

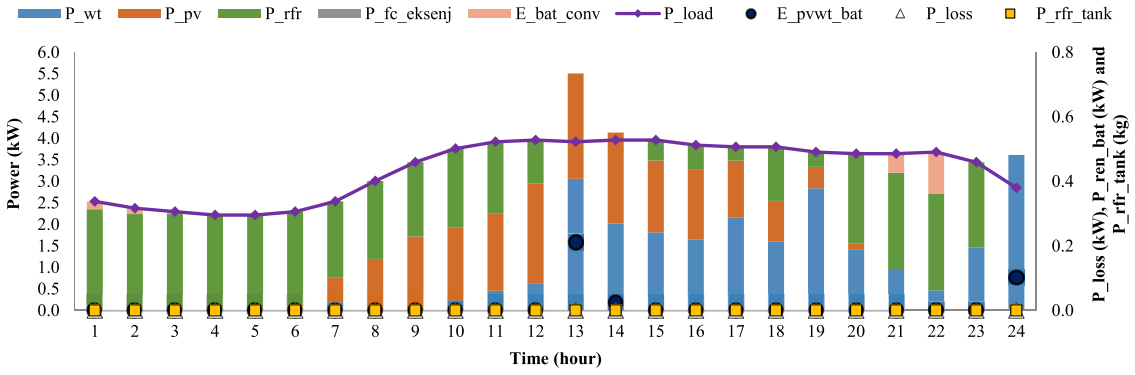


Fig. 20. Energy flow for Kahramanmaraş according to Scenario 2 of a summer day.

demand at hours 13 and 24, and excess energy is directed to batteries for storage. During hours 21 and 22, when energy deficit occurs, the batteries are utilized.

An analysis of the annual energy flows based on monthly averages and the daily energy flows illustrated in Figs. 9–20 reveals that the total energy production from all system components, including the fuel cell and/or batteries, forms columns that align with the fluctuations in the load profile curve. These graphs, which demonstrate the system's adaptability to fluctuations in load demand, confirm the superior performance of the PMS in designing a safe and efficient system.

5.3. Sensitivity analysis

It is essential to undertake a sensitivity analysis to examine the impact of various factors on the overall cost and feasibility of a hybrid energy system dependent on renewable energy [54]. This section systematically investigates the impacts of component cost variations and storage capacities on the economic and operational performance of the system.

5.3.1. The impact of system components' cost variations on the ACS

To assess the sensitivity of the system's economic performance, the Pareto solutions with the highest ACS values in Çanakkale were identified as the most suitable candidates for both Scenario 1 and Scenario 2. To evaluate the individual impacts of each system component on ACS, the capital, operational and maintenance, and replacement costs were sequentially varied during the calculations.

Fig. 21 illustrates the effect of cost variations in all system components on the ACS, the lowest LPSP solution with respect to Scenario 1 for Çanakkale. The graph shows that ACS is most sensitive to changes in fuel

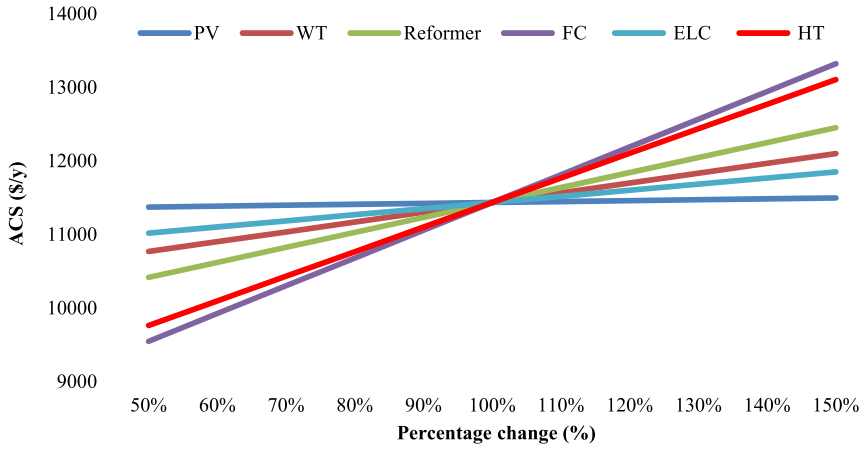


Fig. 21. Sensitivity of ACS to Cost Variations in PV, WT, fuel cell, electrolyzer, and HT (Scenario 1 lowest LPSP solution for Çanakkale).

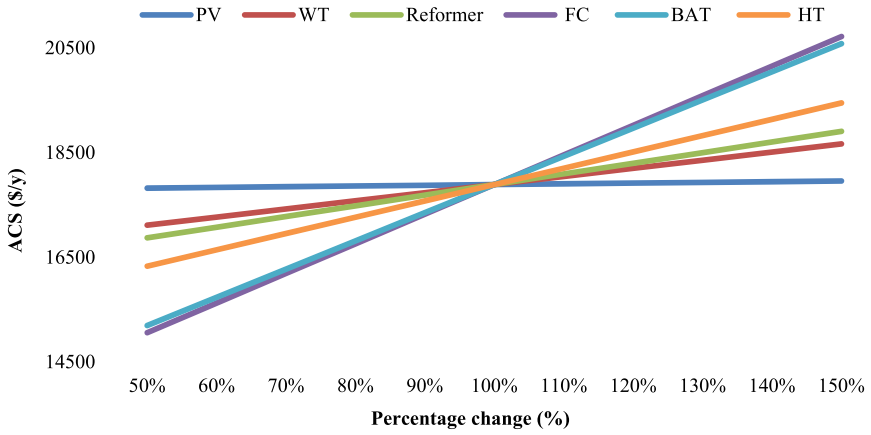


Fig. 22. Sensitivity of ACS to cost variations in PV, WT, fuel cell, battery, and HT (Scenario 2 lowest LPSP solution for Çanakkale).

cell and HT costs. A 50% decrease in fuel cell costs results in a 16.50% reduction in ACS, while a similar decrease in HT costs leads to a 14.6% decline. PV costs have a minimal impact, reducing ACS by only 0.54%. WT, electrolyzer, and reformer costs have moderate effects, decreasing ACS by 5.82%, 8.90%, and 3.63%, respectively.

Fig. 22 illustrates the impact of cost variations across all system components on ACS, focusing on the lowest LPSP solution for Scenario 2 in Çanakkale. Accordingly, ACS is highly sensitive to changes in battery and fuel cell costs; a 50% change in batteries and fuel cells results in a 15.05% and 15.82% change in ACS, respectively. HT costs have a moderate effect, contributing to an 8.72% change in ACS, while reformer, WT, and PV costs lead to changes of 5.69%, 4.34%, and 0.38%, respectively.

5.3.2. *The influence of storage capacities on the objective functions*

Another sensitivity analysis was conducted to evaluate the impact of increasing HT capacity on ACS, LPSP, and PEW, focusing on the lowest ACS solutions for Scenario 1 and Scenario 2 in Çeşme (Solutions 1 in

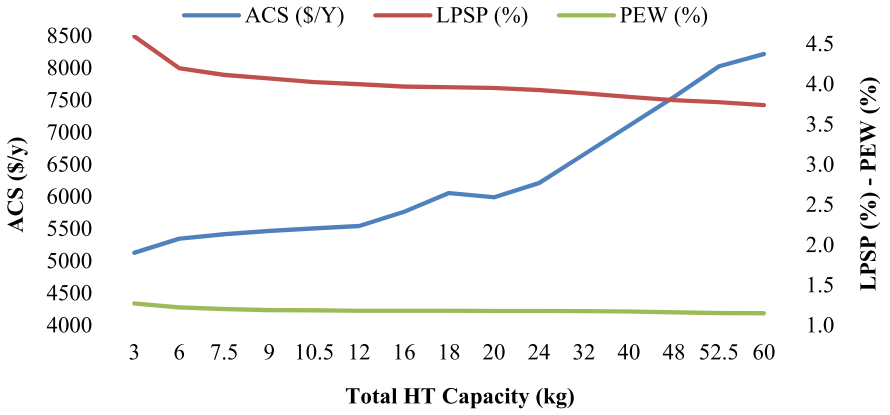


Fig. 23. The influence of increasing total HT capacity on ACS, LPSP, and PEW. (Scenario 1 lowest ACS solution for Çeşme).

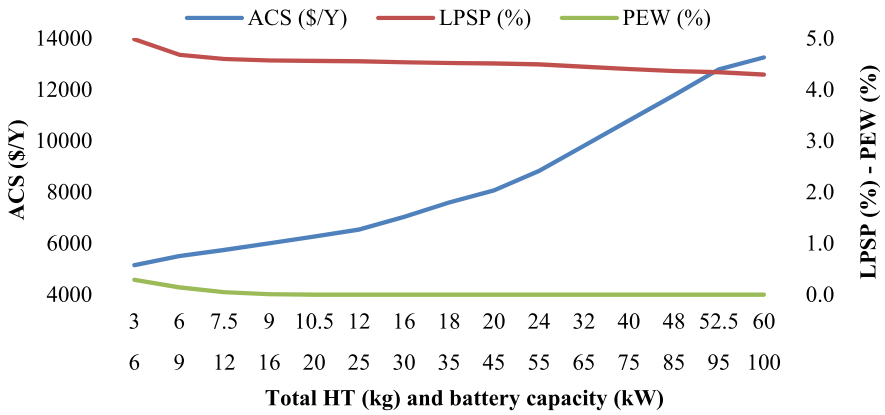


Fig. 24. The impact of increasing HT and battery capacities on ACS, LPSP, and PEW. (Scenario 2 lowest ACS solution for Çeşme).

Tables 11 and 12). For Scenario 1, as illustrated in Fig. 23, the total HT capacity was increased from 3 kg (1.5 kg × 2 units) to 60 kg by adjusting the available size and quantity options. This adjustment resulted in a 60.25% increase in ACS due to higher investment and operational costs. Meanwhile, LPSP decreased by 18.64%, indicating improved reliability, and PEW decreased by 9.54%, indicating better system efficiency. These results emphasize the cost-performance trade-off in optimizing HT capacity.

The analysis depicted in Fig. 24 explores the impact of increasing the total capacities of storage system components on the objective functions for the minimum ACS solution of Scenario 2 in Çeşme. In this context, the total HT capacity was progressively increased from 3 kg (1 kg × 3 units) to 60 kg, and the total battery capacity from 6 kW (3 kW × 2 units) to 200 kW, based on the size and quantity options defined in the sizing algorithm. The findings revealed that while the higher costs of dual storage systems resulted in a 157.35% increase in ACS, they also led to a 13.86% reduction in LPSP and a complete elimination of PEW (100% reduction).

This sensitivity analysis was conducted for the existing lowest ACS solutions for Çeşme by increasing only the storage capacities, while keeping the capacities of all renewable energy sources fixed. Although the LPSP values do not fall below 3% with the current resources, the observed percentage reductions are significant, highlighting the impact of increased storage capacities on improving system reliability. Achieving lower LPSP values would require increasing renewable energy capacities in conjunction with appropriately sized storage systems. This interplay between cost and performance underscores the critical role of multi-objective optimization in developing efficient and reliable hybrid systems.

5.3.3. *The impact of HT and battery capacities on the objective functions*

In the lowest ACS solution of Çeşme Scenario 2 (Table 12, Solution 1), the comparison between hybrid storage and battery-only storage was first analyzed (see Fig. 25). Initially, when the size and quantity of all system components were kept constant, and HT values were set to zero, the changes in ACS, LPSP, and PEW were observed as −5.55%, 40.82%, and 0.00%, respectively. The results highlight that the absence of HT significantly reduces reliability, emphasizing its critical role in mitigating power shortages. Further detailed analysis on the impact of HT absence on LPSP and PEW is provided in Section 5.3.4.

Additionally, the impact of increasing battery capacity in the absence of HT was explored. When the battery capacity was gradually increased, the changes in ACS, LPSP, and PEW shifted to 103.78%, −3.76%, and −100.00%, respectively. As analyzed in Section 5.3.2, hybrid storage significantly impacts cost and reliability. As expected, increasing only the battery capacity resulted in a smaller cost increase compared to hybrid storage (157.35% vs 103.78%). Moreover, the LPSP change in hybrid storage was calculated as −13.86%, while it was limited to −3.76% in battery-only storage. These findings highlight the importance of hybrid storage in improving system reliability.

5.3.4. *The impact of HT and battery presence on LPSP and PEW values*

In Scenario 2 for Çeşme region (23rd solution of the Pareto Front), the LPSP was calculated as 1.33%, and the PEW as 1.02%. To assess the impact of the HT on system performance and reliability, a scenario was examined in which only the battery was used as the storage component. In this case, LPSP increased significantly from 1.33% to 7.26%, while PEW remained constant at 1.02% (see Fig. 26). This stability of the PEW value is due to excess energy being stored solely in batteries instead of

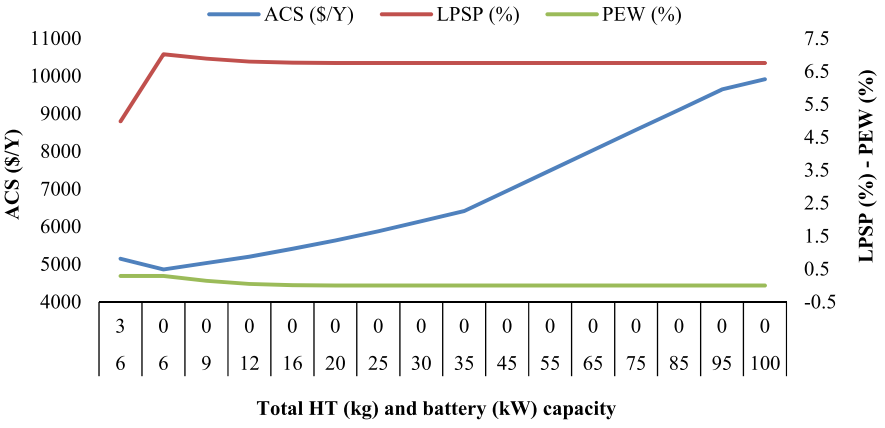


Fig. 25. The impact of HT absence and increased battery capacities on ACS, LPSP, and PEW. (Scenario 2 lowest ACS solution for Çeşme).

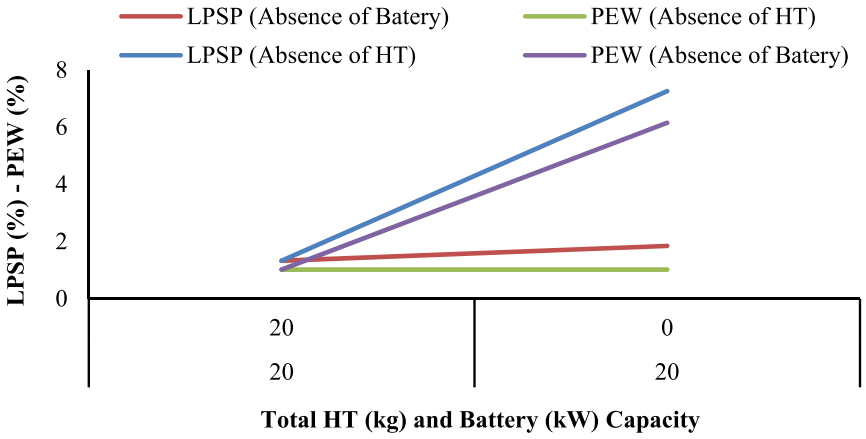


Fig. 26. The impact of the HT and the battery on LPSP and PEW.

being used for hydrogen production via the electrolyzer, along with the battery capacity remaining constant. In the absence of the HT, hydrogen produced by the reformer is directly used to supply the load through the fuel cell. However, when the reformer's hydrogen production exceeds the load demand, the surplus cannot be stored. As a result, during periods of insufficient energy generation, the fuel cell fails to compensate for the energy deficit, leading to a significant increase in LPSP.

In the same Scenario, the impact of the battery on system reliability and efficiency was evaluated by removing the battery from the system. In this case, LPSP increased moderately from 1.33% to 1.85%, while PEW increased significantly from 1.02% to 6.15% (see Fig. 26). The substantial rise in PEW is attributed to the inability to store excess energy generated by PV and WT in the absence of the battery. Conversely, the increase in LPSP was more moderate, as the primary factor ensuring system reliability in this configuration is the continuous hydrogen production enabled by the reformer.

6. Comparison of i-NSGA-II with MOPSO

In this study, the effectiveness of the proposed i-NSGA-II algorithm is demonstrated and validated through a comparative analysis with results obtained using MOPSO algorithm under identical conditions for Çanakkale. Fig. 27 illustrates the Pareto solution sets obtained MOPSO for both Scenario 1 and Scenario 2. In this section, the performance of both optimization techniques is evaluated by comparing the minimum ACS, LPSP, and PEW solutions. Table 16 presents these solutions obtained for Scenario 1 in Çanakkale.

When examining Table 16, it can be observed that the ACS and LPSP values obtained for the minimum ACS solutions using i-NSGA-II are 10% and 19% lower, respectively, while PEW is 2% higher compared to those obtained with MOPSO. Additionally, when considering the total component sizes in the solution, it is noted that i-NSGA-II achieves these superior results with fewer WT and HT compared to MOPSO. These values highlight the optimization performance of the proposed method as well as the superiority of system design optimization, along with the effectiveness of the PMS.

When comparing the minimum LPSP solutions obtained by i-NSGA-II and MOPSO, i-NSGA-II demonstrates a 78% reduction in LPSP, along with a 14% increase in ACS and a slight change in PEW relative to MOPSO. Despite the rise in ACS, the substantial improvement in reliability and the minimal change in PEW highlight i-NSGA-II's ability to provide optimal solutions, resulting in a more reliable system design.

In the evaluation of the minimum PEW solutions, i-NSGA-II demonstrates an 83% decrease in PEW with a 21% and 8% rise in ACS and LPSP, respectively, compared to MOPSO. These results underscore i-NSGA-II's effectiveness in significantly reducing energy waste, leading to a considerable enhancement in system efficiency.

Based on the Pareto optimal solutions presented in Table 17 for Scenario 2, the solutions obtained using i-NSGA-II and MOPSO were evaluated across three different objectives. The minimum ACS solutions achieved a 9% reduction in ACS, while the minimum LPSP solutions resulted in an 85% decrease in LPSP. In the minimum PEW solutions, waste energy was minimized to zero for both approaches, ensuring 100% system efficiency. Moreover, i-NSGA-II outperformed MOPSO

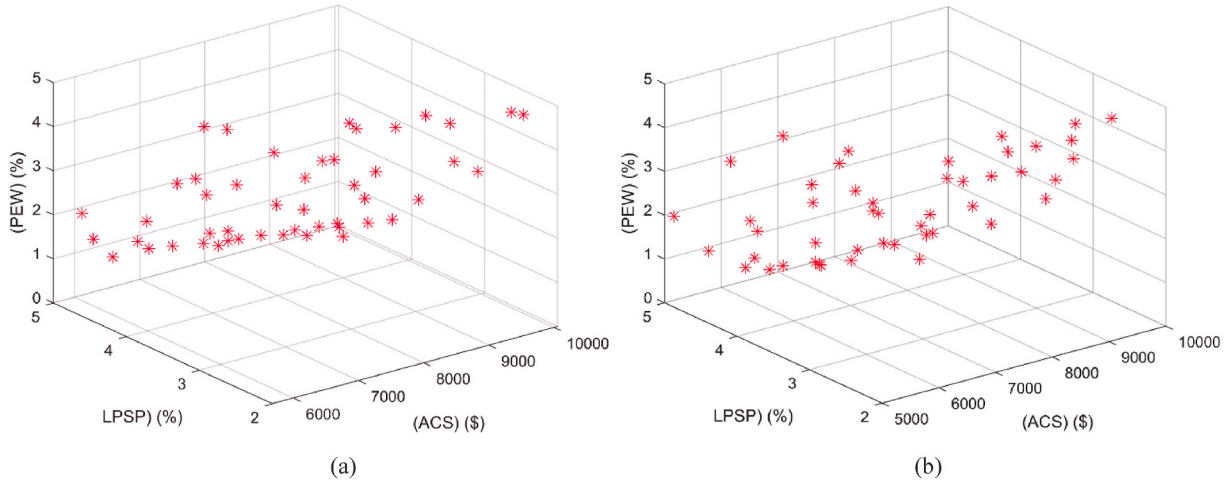


Fig. 27. The Pareto Front obtained with MOPSO for Çanakkale (a) Scenario 1 (b) Scenario 2.

Table 16
Comparison of minimum values of Pareto solutions obtained using i-NSGA-II and MOPSO for Scenario 1 (Çanakkale).

	PV (kW)	WT (kW)	Fuel Cell (kW)	Electrolyzer (kW)	HT (kg)	Reformer (kg)	ACS ($)	LPSP (%)	PEW (%)
i-NSGA-II min ACS Solution	4 (4 kW*1)	3 (1 kW*3)	3 (3 kW*1)	0	4.5 (1.5 kW*3)	0.8	5129.25	4.72	1.89
MOPSO min ACS Solution	4	5	3	0	9	0.8	5674.15	4.61	2.33
i-NSGA-II min LPSP Solution	3 (3 kW*1)	6 (1 kW*6)	6 (3 kW*2)	5 (1 kW*5)	60 (4 kW*15)	0.8	11435.06	0.55	4.47
MOPSO min LPSP Solution	4	10.5	3	5	60	0.8	10056.46	2.47	4.45
i-NSGA-II min PEW Solution	3 (3 kW*1)	4 (4 kW*1)	6 (3 kW*2)	3 (3 kW*1)	60 (4 kW*15)	0.8	10586.14	4.97	0.10
MOPSO min PEW Solution	4	4	3	5	60	0.8	8758.57	4.60	0.61

Table 17
Comparison of minimum values of Pareto solutions obtained using i-NSGA-II and MOPSO for Scenario 2 (Çanakkale).

	PV (kW)	WT (kW)	Fuel Cell (kW)	Battery (kW)	HT (kg)	Reformer (kg)	ACS ($)	LPSP (%)	PEW (%)
i-NSGA-II min ACS Solution	4 (2 kW*2)	3 (1 kW*3)	3 (3 kW*1)	1 (1 kW*1)	4 (4 kW*1)	0.8	5031.17	4.73	1.48
MOPSO min ACS Solution	4	6	3	0	4	0.8	5506.00	4.94	3.34
i-NSGA-II min LPSP Solution	3 (1 kW*3)	7 (1 kW*7)	6 (3 kW*2)	85 (5 kW*17)	60 (4 kW*15)	0.8	15417.91	0.15	4.76
MOPSO min LPSP Solution	5	10.5	3	68	44	0.8	12574.25	0.98	3.98
i-NSGA-II min PEW Solution	2 (1 kW*2)	4 (1 kW*4)	3 (3 kW*1)	40 (5 kW*8)	16 (4 kW*4)	0.8	7945.56	4.03	0.00
MOPSO min PEW Solution	3	6	3	68	9	0.8	9732.64	4.31	0.00

with an 18% improvement in cost and a 6% increase in system efficiency.

These comparisons validate the performance of i-NSGA-II in both scenarios by benchmarking it against MOPSO under identical conditions, offering insights into the trade-offs between system cost, reliability, and efficiency. For all three objective functions, the i-NSGA-II achieved lower values compared to MOPSO, demonstrating its superior performance.

7. Conclusion

This study presents the optimization and analysis of a HRES integrating PVs, WTs, and continuous green hydrogen production via reformers. The system incorporates hybrid storage solutions (HTs and batteries) to enhance reliability and efficiency. The proposed system was optimized using the i-NSGA-II algorithm, integrated with an advanced PMS, and its effectiveness was validated through a comparative analysis with MOPSO, confirming its superior performance. The key findings are as follows.

- The Pareto optimal solutions for hydrogen storage and hybrid storage scenarios in Çanakkale, Çeşme, and Kahramanmaraş were compared. Scenario 2 outperformed Scenario 1 in terms of minimum LPSP and PEW, whereas no significant difference was observed for minimum ACS. Across all locations, the lowest LPSP solutions exhibited a reduction in LPSP values ranging between 74% and 85%, while a 100% reduction was achieved in the minimum PEW solutions. Regarding minimum ACS solutions, in Kahramanmaraş, Scenario 1 provided only a slight advantage of 4% (from 5614.62 kWh to 5821.96 kWh).
- In all three locations, the results indicate that prioritizing one objective function requires compromising on others. Alternatively, a solution that balances all three objectives can be selected from the Pareto front. These findings emphasize that, in multi-objective optimization problems, the selection among conflicting objective functions depends on the preferences of decision-makers.
- The analyses conducted for three different cities have shown that the optimal configuration of HRES is highly influenced by regional conditions.
- The impacts of both regional and seasonal variations, as well as storage options, on system performance have been demonstrated through simulation results. The corresponding graphs show that the total energy output from all system components, including fuel cells and/or batteries, dynamically adapts to fluctuations in the load profile curve. These results confirm the superior performance of the developed PMS in ensuring secure and efficient system operations.
- Sensitivity analyses conducted to measure the impact of system components on ACS revealed that ACS is more sensitive to changes in the total costs of HT and fuel cell in Scenario 1, while in Scenario 2, it is more sensitive to changes in battery and fuel cell costs.
- The analysis of the impact of storage systems on the objective functions revealed that, in both scenarios, increasing the capacities of the existing storage systems raises ACS values while significantly enhancing system reliability and efficiency.
- The sensitivity analyses conducted separately for HT and battery in hybrid storage revealed that HT has a greater impact on system reliability, while the battery plays a more significant role in enhancing system efficiency.
- The sensitivity analysis revealed that variations in costs, storage types, and capacities significantly impact overall system performance.
- Additionally, HRES optimization was performed using MOPSO under the same conditions to validate the effectiveness of the proposed i-NSGA-II. The results confirmed the superior performance of i-NSGA-II, as it achieved better outcomes across all objective functions.

The results demonstrate that the proposed approach provides a robust and flexible framework for optimizing HRES with green hydrogen production and advanced storage solutions, paving the way for future sustainable energy applications.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests.

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